

TYCHE R2-T1

Validation Evidence Compendium

(supporting document to the Final Design Report)

R2-T1 | G1.3 | S1.0 | CVR-1.0

RAeS International Light Aircraft Design Competition 2025–26

Prepared by Glen Mthandeni Ndlovu — Individual Entrant

KPR Institute of Engineering and Technology (KPRIET), Coimbatore, India

30 August 2026

Submission supporting evidence · controlled analytical record

Companion to the analysis archive. Accepted, conditional, rejected and superseded evidence are distinguished throughout; no certification or flight-test approval is implied.

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Reading guide. Section 1 defines the recorded tools and settings; Section 2 states the governing decisions. Sections 3–9 present the technical evidence. Section 10 consolidates open validation items; the appendices index the archive and preserve superseded results.	

1. Document control and analysis environment

1.1 Purpose, scope and evidence status

This compendium organises the Tyche R2-T1 analysis record by engineering question, evidence and disposition. It supports the Final Design Report (FDR) and provides a review route into the associated analysis folders. Rejected models are retained because the acceptance gates, and the decisions not to use invalid coefficients, form part of the verification evidence.

Source basis. The technical tables and recorded run identifiers are drawn from the supplied “Tyche R2-T1 Analysis and Validation Evidence Log”. The latest supplied FDR, “Tyche_R2-T1_Final_Design_Report_G1_3_FINAL_Revised(3).pdf”, is used only to identify the controlling report state. This editorial consolidation does not constitute a new solver execution, independent inspection of the analysis ZIP, or independent validation of supplier data.

Repeated FG8 sections and duplicate coefficient tables have been consolidated. The later atmosphere-consistent VMC calculation supersedes the earlier mixed-density results. Interim CG evidence gaps are distinguished from the later loading-cloud summaries. Original test identifiers and explicitly recorded filenames are retained; missing software versions and archive identifiers are not inferred.

Table 1. Document and configuration control

Item	Controlled statement
Aircraft / geometry	Tyche R2-T1 / G1.3; production fuselage length 10.750 m
Report release tag	R2-T1 G1.3 S1.0 CVR-1.0
Role of this document	Supporting validation evidence; does not replace the self-contained FDR
Evidence precedence	Later explicitly corrected results govern. Diagnostic and historical results are not production coefficients.
Archive identity	Repository URL, release tag, commit identifier and ZIP checksum are not recorded in the supplied log. They belong in the repository release manifest.
Interpretation of PASS	The stated analytical or numerical gate is met on the recorded basis; this is not certification or hardware qualification.

1.2 Tool-and-version register

Table 2. Recorded software and reproducibility information

Tool / record	Version or identifier	Evidence / limitation
OpenVSP	Not recorded in the source log	Geometry, CompGeom, Parasite Drag Tool and control-surface setup. Confirm the exact build from the original environment or archived logs.
VSPAERO	Not recorded in the source log	Thin VLM, thick-panel diagnostics, stability derivatives and powered actuator-disk/free-wake trials. Exact build remains a provenance gap.

Tool / record	Version or identifier	Evidence / limitation
Geometry scripts	G1.4, G1.5 and G1.6 diagnostic revisions	Named .vspscript files are indexed in Appendix A. These revision labels are not software versions or production geometry releases.
Analytical post-processing	Implementation/version not recorded	Munk–Multhopp, CG, steady OEI balance and landing-gear equations are documented here. Source scripts/workbooks must identify the executable calculation basis.
Operating system / runtime	Not recorded	No compiler, OS, Python or spreadsheet version is inferred.
Independent rerun for this issue	Not performed	This issue repackages recorded evidence and checks document consistency.

1.3 Reference geometry and common solver settings

Table 3. Production thin-lifting-system settings

Control	Recorded setting
Representation	MainWing + HTail + VTail + LiftStrutPair; thin VLM; propulsion OFF; no thick body in production coefficients
Reference dimensions	$S = 30.0 \text{ m}^2$; $b = 16.4 \text{ m}$; $\hat{c} = 1.881 \text{ m}$
Baseline moment reference	FS 3.750 m; $Y = Z = 0.000 \text{ m}$. SM6.2 deliberately shifts X to 3.900 m; VMC/FG8 use the stated case references.
Cruise state	66.88 m/s TAS; Mach 0.203; chord Reynolds number 6.98×10^6
Production wake	20 wake iterations; 16 wake nodes; fixed wake OFF; relaxation 1.0; XZ symmetry OFF
Clustering / cap	Leading-edge and trailing-edge clustering 0.25 / 0.25; cap tessellation 3
Load records	218 per bare lifting-system case; 238 when the lift-strut pair is included
Scope of convergence	Case-specific force/moment gates apply. Residual reduction alone is insufficient where symmetry or force credibility fails.

Table 4. Surface-discretisation register

Level	MainWing W; section U	HTail W / U	VTail W / U
Coarse	33; 3 / 11 / 3 / 7	21 / 9	21 / 9
Medium	33; 5 / 17 / 3 / 9	33 / 13	33 / 13
Fine	49; 7 / 25 / 5 / 13	49 / 19	49 / 19

Level	MainWing W; section U	HTail W / U	VTail W / U
Extra-Fine (production)	73; 9 / 37 / 7 / 19	73 / 29	73 / 29

W and U retain the source's OpenVSP tessellation labels. The executed G1.5/G1.6 diagnostic models were restored to Extra-Fine settings; lower defaults in a script are not evidence of the settings actually run. The clean convergence campaign also evaluated 16 versus 20 wake iterations and 8 versus 16 wake nodes.

Table 5. Campaign-specific departures from the common settings

Campaign	Flow / configuration	Solver / interpretation
CA-G0/G1/G3	-4° to +10° angle-of-attack sweep; cruise state	Extra-Fine; 20 iterations; 16 nodes; attached-flow clean polar
G5 derivatives	Cruise state; manual sideslip -6° to +6°; small control deflections	Production thin model; zero-sideslip numerical intercepts retained
VMC7.1 / VMC7.2	20.58 m/s; Mach 0.0605; Re = 2.65×10^6 ; density 1.225 kg/m ³	Unpowered low-speed configuration and +12° Fowler checks
VMC7.3C-F	23.60 m/s; Mach 0.0693; Re = 3.04×10^6 ; density 1.225 kg/m ³	40 iterations; 16 nodes; relaxation 1.0 / 0.5 / 0.5 / 0.25; powered diagnostics rejected
FG8.1	1,608.8 kg; FS 3.37374 m; 22.16 m/s; Mach 0.0634; Re = 2.41×10^6	Density 1.079 kg/m ³ ; +12° Fowler; propulsion OFF
FG8.2 / FG8.3	1,519.8 kg; FS 3.35170 m; 21.72 m/s; Mach 0.0621; Re = 2.35×10^6	Density 1.079 kg/m ³ ; +35° / +20° / +12° Fowler control surrogates; propulsion OFF

FG8 provenance note. Early positive-elevator files contain “M” in the filename but record positive GUI commands; filenames alone do not establish elevator sense. Early unpowered files used a 1.225 kg/m³ setting while dimensional loads were evaluated at 1.079 kg/m³; the source retained their non-dimensional coefficients with imposed Reynolds number. The later nose-up cases use the corrected hot-high state. These distinctions must remain visible in the archive.

1.4 Conventions and numerical acceptance

Table 6. Coefficient mapping

Conventional quantity	VSPAERO history field	Mapping
Lift coefficient, C_L	CLtot	Direct
Wake lift coefficient, $C_{L,w}$	CLwtot	Direct
Induced drag, C_{Di}	CDi	Direct
Wake induced drag, $C_{Di,w}$	$C_{Di,w}$	Direct
Side-force coefficient, C_Y	CStot	Direct

Conventional quantity	VSPAERO history field	Mapping
Rolling-moment coefficient, C_l	CMxtot	$C_l = -C M x t o t$
Pitching-moment coefficient, C_m	CMytot	Direct
Yawing-moment coefficient, C_n	CMztot	$C_n = -C M z t o t$

$$Span_{5,\%}(X) = 100 \frac{\max(X_{N-4:N}) - \min(X_{N-4:N})}{|\bar{X}_{N-4:N}|} \quad (1)$$

The final-five relative span is used only where its denominator is meaningful; near-zero lateral coefficients and pitching moment use absolute-span gates. Steady-rate derivatives are non-dimensional rate derivatives; static and control derivatives are per radian unless otherwise stated. All angles in the equilibrium equations are in radians; tables may report degrees.

Table 7. Preferred low-speed and powered numerical gates

Quantity	Preferred numerical gate
C_L , $C_{L,w}$, $C_{D_i,w}$ final-five relative span	less than 0.5%
C_m final-five absolute span	less than 0.005
C_Y , C_l , C_n final-five absolute span	less than 5×10^{-4} preferred
Residuals	Sustained decay without late oscillation or sign cycling

Acceptance is conjunctive: the intended geometry and flow state, numerical stability, physical symmetry, force/moment credibility and the intended use must all be satisfactory. A solver result that converges to a physically invalid state is rejected. A configuration check is not automatically a validated drag polar or stall prediction.

2. Governing evidence and decision ledger

2.1 Controlled conclusions

Table 8. Governing evidence summary

Evidence block	Retained result	Disposition / boundary
CompGeom / PDT	Complete-aircraft wetted area 141.591 m ² ; geometry-only parasite drag 0.018985	Accepted build-up; operational and gear model-form allowances remain explicit
Clean aerodynamics	Extra-Fine/16 thin lifting-system CA-G3 polar	Accepted with residual induced-drag mesh sensitivity; mixed-body coefficients excluded
G5 static / control derivatives	Converged manual sideslip and control sweeps; favourable direct damping signs	Accepted for stated linear screens; not a complete coupled-mode solution
SM6 fuselage / CG	Central corrected NP FS 4.02894 m; governing U20 aft screen FS 3.83778 m	Classical conceptual correction; CAD-quality arms and complete-aircraft model validation remain open
VMC7.4 corrected atmosphere	Sea-level VMC-A: 38.97 kn EAS windmill; 39.37 kn severe	Physical-stop screen passes narrowly; 40.97 kn at 90% travel fails robustness
FG8 longitudinal control	Approximately 15° nose-up rotation; converged 20° authority; landing trim approximately 4.08° nose-up	Quasi-steady unpowered control screen only; actual high-lift/flare dynamics not validated
FG8.4 gear	Each-main demand 33.61 / 50.41 kN limit / ultimate	Existing design basis 44.2 / 66.3 kN retained; nose hardware and ground handling open

2.2 Model selection, rejection and supersession

Table 9. Decision ledger

ID / branch	Decisive evidence	Decision and consequence
DL-01 · G2 strut attachment	Root trimming recovered; final area 141.591 m ² versus interim 141.533 m ²	Corrected G2 selected. Interim geometry is historical, not the production area.
DL-02 · surface/wake refinement	Lift passes; fitted induced-drag factor remains about 5.25% mesh-sensitive	Extra-Fine/16 retained with uncertainty; no claim of complete induced-drag mesh convergence.
DL-03 · mixed thin/thick body	Root defects repaired, but combined solutions retain spurious zero-sideslip force and inconsistent forces/residuals	Rejected for production. Use thin lifting system, separate PDT and analytical fuselage correction.

ID / branch	Decisive evidence	Decision and consequence
DL-04 · low-speed flap grouping	Original group produced large asymmetric roll; corrected grouping restored symmetric flap response	Initial grouping rejected; corrected VMC7.2 configuration retained.
DL-05 · initial actuator disk	Thrust/power coefficients imply propulsive efficiency $1.667 > 1$	Physically impossible input pair rejected; corrected input pair retained only for diagnostics.
DL-06 · powered free wake	C–F cases fail final-five/residual gates despite counter-rotation and reduced relaxation	Powered coefficients rejected. VMC7.4 uses a bounded analytical OEI balance instead.
DL-07 · reference and CG closure	SM6.2 was subsequently executed; body correction makes FS 3.900 m fail the 10% margin screen	Earlier “skip audit” and uncorrected aft-boundary conclusions superseded. U20 screen governs.
DL-08 · VMC atmosphere	Earlier thrust used ambiguously traced density 1.168 kg/m^3	Explicit sea-level and hot-high calculations supersede 38.73 / 39.09 kn headlines.
DL-09 · duplicate FG8 issue	The later issue repeats FG8.0–3 and adds FG8.4 plus atmosphere correction	One integrated FG8 record retained; duplicate diary sections removed.
DL-10 · validation limits	No complete hardware, high-lift flight correlation or coupled-mode evidence in this log	Open items remain explicit; no new evidence is implied by editorial closure.

2.3 Relationship to the final report state

The analysis cases retain the masses, references and atmospheres actually recorded. They have not been silently rebased to later fuel or tank-system changes. The FDR’s current mass/fuel values are included below for configuration reconciliation, not as a new calculation campaign.

The current FDR gives OEW 1,329.0 kg, nominal dispatch fuel 134.9 kg and full-payload mission mass 2,179.9 kg, exceeding the 2,000 kg MTOM by 179.9 kg. The 190 L gross installation gives approximately 514.8 km nominal range; the upper-flow 201.2 L requirement remains open.

Historical CG and FG8 cases retain their executed masses. The later 2.0 kg tank-system increment requires incorporation at its ledger arm before operational CG release.

Installed continuous power is 196.6 kW against 169.0 kW cruise demand; the VMC high-thrust bound is a separate operating state. The FDR also carries bounded phugoid, roll-subsidence and short-period screens whose detailed derivations are not in this uploaded log. Dutch-roll and spiral closure remain HOLD. No aerodynamic or control PASS removes the aircraft’s mass shortfall.

3. Geometry and parasite-drag evidence

3.1 CompGeom configuration build-up

The accepted geometry progresses from bare airframe G0 to nacelle-installed G1 and the corrected complete unpowered G2. Symmetric MainWing and HTail mesh merges are recorded and accepted. No new warning accompanies the corrected strut/gear installation.

Table 10. Accepted G0–G2 CompGeom progression

Metric	G0 — Bare Airframe	G1 — Nacelle Airframe	G2 — Complete Airframe	Assessment
Geoms	4	6	11	Correct progression
Meshes	6	8	16	PASS
Triangles	8,288	9,824	13,888	Appropriate growth
Fuselage (Swet)	44.529	44.529	44.483	G2 attachment trimming
Wing (Swet)	59.599	58.530	58.488	Nacelle + strut trimming
HT (Swet)	16.074	16.074	16.074	Stable
VT (Swet)	7.917	7.917	7.917	Stable
Nacelles	—	7.436	7.436	Stable
Lift struts	—	—	4.260	Installed exposed area
Main gear legs	—	—	0.972	Installed exposed area
Main wheels	—	—	1.362	Installed exposed area
Nose gear leg	—	—	0.137	Installed exposed area
Nose wheel	—	—	0.460	Installed exposed area
Total (Swet)	128.118	134.486	141.591	PASS
MainWing merge warning	Yes	Yes	Yes	Accepted
HTail merge warning	Yes	Yes	Yes	Accepted
Other warnings	None	None	None	PASS

Metric	G0 — Bare Airframe	G1 — Nacelle Airframe	G2 — Complete Airframe	Assessment
Overall status	PASS	PASS	PASS	CompGeom gate closed

Each nacelle contributes 3.718 m² exposed area. Nacelle/wing trimming removes 1.069 m² of main-wing area, giving a net installed nacelle increment of 6.368 m². In corrected G2, each lift strut has 2.223 m² theoretical and 2.130 m² exposed area; main-wing and fuselage reductions of 0.042 and 0.046 m² provide a consistent attachment-trimming check. The earlier 141.533 m² G2 result is retained only in the supersession ledger.

Table 11. Geometry-set release status

Set	Purpose	Final wetted area	Status
G0_BARE_AIRFRAME	Wing-body-tail baseline	128.118 m ²	PASS
G1_NACELLE_AIRFRAME	G0 + installed nacelles	134.486 m ²	PASS
G2_COMPLETE_AIRFRAME	G1 + struts + fixed gear	141.591 m ²	PASS
G3_POWERED_AIRCRAFT	Adds actuator-disk props	Not required for wetted-area CompGeom	N/A

3.2 Component parasite-drag build-up

The OpenVSP Parasite Drag Tool (PDT) provides the complete-aircraft zero-lift build-up. The recorded cruise basis is 66.88 m/s TAS (130 kn), 2,600 m standard atmosphere and reference area 30.0 m²; the components are treated as fully turbulent. VSPAERO's incomplete lifting-model zero-lift drag is not substituted for this build-up.

$$C_{D0} = \frac{f}{S}, f = \sum_i C_{f,i} F F_i Q_i S_{wet,i} \quad (2)$$

Table 12. G0 bare-airframe PDT inputs and outputs

Parameter	Fuselage	Main Wing	H-Tail	V-Tail	Total
(Swet), m ²	44.5289	59.5985	16.0743	7.9167	128.1184
L_{ref} , m	10.8061	1.8805	1.4028	1.5080	—
(t/c) or body ratio	6.0092	0.1701	0.1199	0.1199	—
Reynolds number	4.011×10^7	6.980×10^6	5.207×10^6	5.597×10^6	—
Laminar fraction	0%	0%	0%	0%	—
C_f	0.002422	0.003121	0.003267	0.003230	—

Parameter	Fuselage	Main Wing	H-Tail	V-Tail	Total
Form factor, (FF)	1.1341	1.3903	1.2521	1.2521	—
Interference (Q)	1.000	1.000	1.000	1.000	—
Flat-plate area (f), m ²	0.12233	0.25860	0.06576	0.03202	0.47871
C_{D0}	0.004078	0.008620	0.002192	0.001067	0.015957
Share of G0 drag	25.55%	54.02%	13.74%	6.69%	100%
Status	PASS	PASS	PASS	PASS	PASS

Table 13. G1 nacelle-installed PDT inputs and outputs — panel A

Parameter	Fuselage	Main Wing	H-Tail	V-Tail
(Swet), m ²	44.529	58.530	16.074	7.917
L_{ref} , m	10.806	1.881	1.403	1.508
(t/c) or fineness ratio	6.009	0.170	0.120	0.120
Reynolds number	4.011×10^7	6.980×10^6	5.207×10^6	5.597×10^6
Laminar fraction	0%	0%	0%	0%
C_f	0.002422	0.003121	0.003267	0.003230
Form factor (FF)	1.134	1.390	1.252	1.252
(Q)	1.00	1.00	1.00	1.00
Flat-plate area (f), m ²	0.122329	0.253967	0.065758	0.032019
C_{D0}	0.004078	0.008466	0.002192	0.001067
% of G1 total	24.08%	50.00%	12.95%	6.30%
Status	PASS	PASS	PASS	PASS

Table 14. G1 nacelle-installed PDT inputs and outputs — panel B

Parameter	Nacelle R	Nacelle L	Total
(Swet), m ²	3.718	3.718	134.486
L_{ref} , m	2.200	2.200	—
(t/c) or fineness ratio	3.142	3.142	—
Reynolds number	8.165×10^6	8.165×10^6	—
Laminar fraction	0%	0%	—

Parameter	Nacelle R	Nacelle L	Total
C_f	0.003047	0.003047	—
Form factor (FF)	1.495	1.495	—
(Q)	1.00	1.00	—
Flat-plate area (f), m ²	0.016934	0.016934	0.507942
C_{D0}	0.000564	0.000564	0.016931
% of G1 total	3.33%	3.33%	100%
Status	PASS	PASS	PASS

Table 15. G2 complete-aircraft component PDT build-up — panel A

Component	S_{wet} (m ²)	L_{ref} (m)	Re
Fuselage	44.483	10.806	4.01×10^7
Main wing	58.488	1.881	6.98×10^6
H-tail	16.074	1.403	5.21×10^6
V-tail	7.917	1.508	5.60×10^6
Nacelle R	3.718	2.200	8.17×10^6
Nacelle L	3.718	2.200	8.17×10^6
Lift struts, pair	4.261	0.285	1.06×10^6
Main gear legs, pair	0.973	0.211	7.84×10^5
Main wheels, pair	1.363	0.600	2.23×10^6
Nose gear leg	0.137	0.201	7.45×10^5
Nose wheel	0.460	0.500	1.86×10^6
TOTAL	141.591	—	—

Table 16. G2 complete-aircraft component PDT build-up — panel B

Component	C_f	FF	f (m ²)
Fuselage	0.002422	1.134	0.122204
Main wing	0.003121	1.390	0.253784
H-tail	0.003267	1.252	0.065758
V-tail	0.003230	1.252	0.032019

Component	C_f	FF	f (m ²)
Nacelle R	0.003047	1.495	0.016934
Nacelle L	0.003047	1.495	0.016934
Lift struts, pair	0.004278	1.367	0.024921
Main gear legs, pair	0.004518	1.622	0.007129
Main wheels, pair	0.003754	4.228	0.021629
Nose gear leg	0.004562	1.576	0.000983
Nose wheel	0.003873	4.078	0.007267
TOTAL	—	—	0.569563

Table 17. G2 complete-aircraft component PDT build-up — panel C

Component	C_{D0}	Share	Assessment
Fuselage	0.004073	21.46%	PASS
Main wing	0.008459	44.56%	PASS
H-tail	0.002192	11.55%	PASS
V-tail	0.001067	5.62%	PASS
Nacelle R	0.000564	2.97%	PASS
Nacelle L	0.000564	2.97%	PASS
Lift struts, pair	0.000831	4.38%	PASS
Main gear legs, pair	0.000238	1.25%	PASS/REVIEW
Main wheels, pair	0.000721	3.80%	REVIEW
Nose gear leg	0.000033	0.17%	PASS/REVIEW
Nose wheel	0.000242	1.28%	REVIEW
TOTAL	0.018985	100%	PASS as baseline

Table 18. Configuration-level parasite-drag reconciliation

Metric	G0 Bare	G1 + Nacelles	G2 Complete	Increment
S_{wet} , m ²	128.118	134.486	141.591	+13.473 vs G0
Flat-plate area (f), m ²	0.478708	0.507942	0.569563	+0.090855
C_{D0}	0.015957	0.016931	0.018985	+0.003028 vs G0

Metric	G0 Bare	G1 + Nacelles	G2 Complete	Increment
Increase from previous set	—	+6.10%	+12.13%	—
Increase from G0	—	+6.10%	+18.98%	—
Status	PASS	PASS	PASS baseline	—

3.3 Operational and landing-gear sensitivities

Table 19. PDT operational, cooling and gear model-form cases — panel A

Case	Purpose	Operational / cooling allowance	Gear treatment	Total (f) (m ²)
S0	Raw geometry baseline	0%	1.0×	0.569563
S1	Low operational sensitivity	+5% of $C_{D,geom}$	1.0×	0.598041
S2	Moderate operational sensitivity	+10% of $C_{D,geom}$	1.0×	0.626519
S3	High operational sensitivity	+15% of $C_{D,geom}$	1.0×	0.654998
G1	Gear-model sensitivity	0%	1.5× effective gear drag	0.588073
G2	Strong gear sensitivity	0%	2.0× effective gear drag	0.606583
C1	Conservative working case	+10%	1.5× gear	0.645029
C2	Upper-bound sensitivity	+15%	2.0× gear	0.692018
Report performance basis	Legacy analytical estimate	—	—	0.849000 equivalent

Table 20. PDT operational, cooling and gear model-form cases — panel B

Case	Total C_{D0}	Increase vs S0	Difference from 0.0283
S0	0.018985	—	−32.92%
S1	0.019935	+5.00%	−29.56%
S2	0.020884	+10.00%	−26.20%

Case	Total C_{D0}	Increase vs S0	Difference from 0.0283
S3	0.021833	+15.00%	-22.85%
G1	0.019602	+3.25%	-30.73%
G2	0.020219	+6.50%	-28.55%
C1	0.021501	+13.25%	-24.02%
C2	0.023067	+21.50%	-18.49%
Report performance basis	0.028300	+49.1% vs S0	—

The geometry-only result is 0.018985; the conservative operational case is 0.021501 and the deliberately adverse upper build-up is 0.023067. The FDR retains 0.0283 as a conservative performance assumption, not a directly validated PDT result. It is approximately 22.7% above the executed upper build-up; equivalently, that upper build-up is 18.5% below the performance assumption. The validation model is not tuned to force agreement.

4. Numerical convergence and clean aerodynamics

4.1 Surface, wake and iteration convergence

The mesh study progressed through 62, 94, 144 and 218 lifting-load records. Initial 16-iteration and eight-node trials identified sensitivity in induced drag and pitching moment. The retained production basis is 20 iterations with 16 wake nodes; the two-by-two comparison separates surface from wake effects.

Table 21. Two-by-two surface/wake comparison

Quantity	Fine / 8	Fine / 16	Extra-Fine / 8	Extra-Fine / 16
$C_L(4^\circ)$	0.819832	0.814470	0.823162	0.820990
$C_{D_i}(4^\circ)$	0.027206	0.027961	0.029758	0.030046
$C_{M_y}(4^\circ)$	-0.011634	-0.003911	-0.001913	-0.004905
$C_{L\alpha}$, rad^{-1}	5.6121	5.5667	5.5671	5.5503
low-angle $C_{m\alpha}$, rad^{-1}	-1.3443	-1.2689	-1.3225	-1.3452
fitted k	0.039071	0.040751	0.042470	0.043010
fitted e	0.9087	0.8713	0.8360	0.8255
VSPAERO (E) at 4°	0.8771	0.8423	0.8085	0.7965

Table 22. Final clean convergence assessment

Verification metric	Fine / 16	Extra-Fine / 16	Difference	Criterion	Disposition
$C_L(4^\circ)$	0.81447	0.82099	0.79%	$\leq 1\%$	PASS
$C_{L\alpha}$	5.5667	5.5503	0.30%	$\leq 2\%$	PASS
$C_{D_i}(4^\circ)$	0.027961	0.030046	6.94%	$\leq 2\%$	OPEN
fitted k	0.040751	0.043010	5.25%	$\leq 2\%$	OPEN
fitted e	0.8713	0.8255	5.54%	$\approx \leq 2\%$	OPEN
$C_{M_y}(4^\circ)$	-0.003911	-0.004905	0.000993 abs.	≤ 0.005	PASS
low-angle $C_{m\alpha}$	-1.2689	-1.3452	5.67%	$\leq 2\%$	OPEN
Wake-node sensitivity in k	4.12%	1.26%	—	$\leq 2\%$	XF PASS
Iteration convergence	PASS	PASS	—	stable	PASS

Table 23. Extra-Fine / 16 final wake-iteration changes

α	$\Delta C_L, 19 \rightarrow 20$	$\Delta C_{D_i}, 19 \rightarrow 20$	$\Delta C_{M_y}, \text{absolute}$
-2°	+0.015%	+0.039%	-0.000126
0°	+0.011%	-0.092%	+0.000013
2°	-0.039%	-0.192%	+0.000226
4°	+0.095%	+0.066%	-0.000240

Fine-to-Extra-Fine differences of 0.79% in lift at 4° and 0.30% in lift-curve slope meet the adopted 1% / 2% criteria. The surface induced-drag coefficient remains 6.94% sensitive at 4° , and fitted k remains 5.25% sensitive. The low-angle pitching-moment slope has a 5.67% mesh difference even though the local moment-coefficient difference passes. These residuals remain uncertainty terms; “accepted” does not mean every numerical target is closed.

4.2 No-strut baseline and strut sensitivity

Table 24. CA-G0 clean lifting-system polar

α	C_L	C_{D_i}	C_{M_y}	VSPAERO E
-4°	0.04480	0.001740	+0.17335	0.0410*
-2°	0.23968	0.003585	+0.13629	0.5690
0°	0.43405	0.008912	+0.09512	0.7505
2°	0.62753	0.017704	+0.04917	0.7897
4°	0.82098	0.030046	-0.00487	0.7965
6°	1.01467	0.045806	-0.06429	0.7980
8°	1.20781	0.065221	-0.13566	0.7941
10°	1.39983	0.087932	-0.21194	0.7912

The near-zero-lift efficiency value at -4° is ill-conditioned and is not interpreted physically. CA-G0 final-iteration maxima are 0.112% in lift, 0.234% in induced drag and 0.00113 absolute in pitching moment. The fitted lift-curve slope is 5.546 rad^{-1} , with lift intercept 0.4337 and zero-lift angle -4.48° ; the positive-lift fit gives $k = 0.04467$ and e approximately 0.795.

Table 25. CA-G1 lift-strut sensitivity polar

α	C_L	C_{D_i}	C_{M_y}	VSPAERO E
-4°	0.03992	0.001796	+0.16825	0.0308*
-2°	0.23710	0.003570	+0.13364	0.5583
0°	0.43388	0.008909	+0.09500	0.7508
2°	0.62981	0.017802	+0.05172	0.7926
4°	0.82519	0.030283	+0.00207	0.8005

α	C_L	C_{D_i}	C_{M_y}	VSPAERO E
6°	1.02157	0.046410	-0.05456	0.8012
8°	1.21774	0.066224	-0.12438	0.7985
10°	1.41107	0.089222	-0.19537	0.7966

Adding the lift-strut pair changes lift and surface induced drag at 4° by +0.51% and +0.79%, respectively. The fitted induced-drag factor changes by -0.12%. The principal effect is pitching moment: the slope changes from -1.345 to -1.255 rad⁻¹, a 6.7% reduction in stabilising magnitude. Symmetry and convergence remain acceptable.

4.3 CA-G3 production polar and repeatability

Table 26. Final CA-G3 clean polar

α	C_L	C_{D_i} surface	$C_{D_{i,w}}$ wake	C_{M_y}	VSPAERO E
-4°	0.039914	0.001796	0.001425	+0.168266	0.0308*
-2°	0.237099	0.003570	0.002456	+0.133656	0.5583
0°	0.433879	0.008909	0.006315	+0.095006	0.7508
2°	0.629810	0.017802	0.013043	+0.051722	0.7926
4°	0.825266	0.030292	0.022624	+0.001833	0.8004
6°	1.021521	0.046402	0.035128	-0.054491	0.8012
8°	1.217688	0.066223	0.050645	-0.124256	0.7985
10°	1.411015	0.089204	0.069232	-0.194975	0.7967

CA-G3 reproduces the accepted CA-G1 4° solution within 0.017% in lift, 0.055% in surface induced drag and 0.00021 absolute in pitching moment. The recorded production 4° coefficients are 0.825049 lift, 0.0302665 surface induced drag and +0.002280 pitching moment; wake lift and drag are 0.826879 and 0.0226181. Surface and Trefftz values are not interchangeable.

$$C_{D,nom} = 0.0216437 + 0.0446077 C_L^2 \quad (3)$$

$$C_{D,adv} = 0.0232097 + 0.0446077 C_L^2 \quad (4)$$

$$C_{D,combinedadv} = 0.0232097 + 0.0469719 C_L^2 \quad (5)$$

These are the source's clean sensitivity fits, formed from PDT allowances and the surface induced-drag fit. The nominal and combined-adverse maximum lift-to-drag ratios are approximately 16.09 and 15.14. They do not replace the FDR's conservative performance polar. The source's final register also contained a generic instruction to use Trefftz drag; that wording is not applied retrospectively to a surface-drag fit. Wake drag remains an independent diagnostic and convergence cross-check.

Table 27. Direct nominal and adverse clean-polar results — panel A

α	C_L	C_{D_i}	$C_{D,nom}$
-4°	0.0399	0.00180	0.02330
-2°	0.2371	0.00357	0.02507
0°	0.4339	0.00891	0.03041
2°	0.6298	0.01780	0.03930
4°	0.8253	0.03029	0.05179
6°	1.0215	0.04640	0.06790
8°	1.2177	0.06622	0.08772
10°	1.4110	0.08920	0.11071

Table 28. Direct nominal and adverse clean-polar results — panel B

α	$C_{D,adv}$	$(L/D)_{nom}$	$(L/D)_{adv}$
-4°	0.02486	1.71	1.61
-2°	0.02664	9.46	8.90
0°	0.03198	14.27	13.57
2°	0.04087	16.02	15.41
4°	0.05336	15.93	15.47
6°	0.06947	15.04	14.70
8°	0.08929	13.88	13.64
10°	0.11227	12.75	12.57

4.4 Mixed thin/thick body branch: rejection ledger

The body-interference investigation tested isolated thick fuselage and wing models, mixed thin/thick assemblies, all-thick combinations and controlled root repairs. The isolated thick MainWing at 4° converged to lift 0.9094, induced drag 0.03496 and pitching moment -0.01467 , with final-step changes 0.001%, 0.007% and 2.1×10^{-6} . Combining components nevertheless introduced nonphysical junction loads. Isolated success was therefore not promoted as full-aircraft validation.

Table 29. Geometry-repair decisions

Branch / test	Observed mechanism	Disposition
Initial CA-G2 junction trials	Depending on tessellation, a clipped 20–31 mm root sliver or an oversized 5.33 m root load polygon; $W = 73 / 49 / 33$ and fuselage $U = 6 / 9$ did not resolve combined physics	Rejected mixed and all-thick combined coefficients
G1.4 MainWing Mode 2	First exposed root chord approximately 2.270 m; root boundary near $y = \pm 0.6967$ m	AC-02 local topology repair accepted; not a production geometry release
G1.4 H2 / G1.5 fairing	Adding tails introduced approximately 0.103 m remnant root strips; sliver removal did not cure one-sided loading	Combined branch rejected
G1.5 isolated HT_TEST	Strong residual convergence but incorrect lift sense, negative induced drag and spurious side force	Rejected on physical-validity grounds
G1.5 direct tail penetration	60 mm tail-cone rise improved isolated behaviour; combined H2 still failed	Diagnostic only
G1.6 0.600 m fairing	First exposed strip only about 28% of normal chord	Topology not accepted
G1.6 0.680 m fairing	Trim boundary $y = \pm 0.3400$ m; first exposed chord 1.63110 m, approximately 92.1% of full chord	AC-04 local repair accepted; combined model not accepted
G1.6 HT_TEST_4	Residual $\log_{10} L_2 = -3.917$ but zero-sideslip side force approximately +0.07628	Numerically converged, physically invalid; rejected
Final extended aft fairing	Longer taper reduced local pressure peak to approximately 0.325, while combined side force remained approximately +0.05441	Campaign closed; no mixed-body production coefficients

Table 30. G1.4–G1.5 diagnostic result ledger — panel A

Test	Primary CSV	C_L	$C_{D_{t,w}}$	C_{M_y}
G1.4-H1-M1	TRIAL	0.6710	0.01753	-0.0089
G1.4-H1-M2	H1_MODE2	0.6897	0.01846	-0.0192
G1.4-H2	H2_Mode_1	0.3912	0.01697	+0.8524
G1.5-H2-TRF	H2_Mode_1(1)	0.5993	0.01810	+0.2622

Test	Primary CSV	C_L	$C_{D_i,w}$	C_{M_y}
G1.5-HT-TRF	HT_TEST	-0.0302	0.000760	+0.0951
G1.5-HT-RISE060	HT_Test_2	0.0460	0.000695	-0.1385
G1.5-H2-RISE060	H2_Test_2	0.6579	0.01822	+0.0750
G1.5-H2-NOVT	H2_NO_VT_TEST	0.5413	0.01717	+0.4256

Table 31. G1.4–G1.5 diagnostic result ledger — panel B

Test	CS	log10 L2	Disposition
G1.4-H1-M1	+0.00008	-0.145	Conditional
G1.4-H1-M2	-0.00055	+0.265	Retain AC-02
G1.4-H2	+0.01766	+0.167	Fail
G1.5-H2-TRF	+0.00459	-0.295	Fail
G1.5-HT-TRF	-0.00740	-3.841	Fail physical
G1.5-HT-RISE060	-0.00014	-3.392	Pass diagnostic
G1.5-H2-RISE060	+0.00923	+0.784	Fail
G1.5-H2-NOVT	-0.00267	+0.714	Fail

Table 32. G1.6 diagnostic result ledger — panel A

Test	Primary CSV	C_L	$C_{D_i,w}$	C_{M_y}
G1.6-HT1	HT_TEST_1	0.1058	0.000659	-0.0756
G1.6-HT2	HT_TEST_2(1)	0.0449	0.000818	-0.1310
G1.6-HT3	HT_TEST_3	0.0452	0.000796	-0.1319
G1.6-HT4	HT_TEST_4	0.1086	0.000663	-0.0789
G1.6-H2-NOVT	H2_NO_VT_TEST_1	0.6975	0.01803	+0.0736
G1.6-H2-LONG	H2_TEST_1	0.7032	0.01834	+0.0545

Table 33. G1.6 diagnostic result ledger — panel B

Test	CS	log10 L2	Disposition
G1.6-HT1	+0.07662	-1.072	Hold/Fail
G1.6-HT2	+0.00142	-0.721	Hold
G1.6-HT3	-0.00036	-3.696	Pass diagnostic

Test	CS	log10 L2	Disposition
G1.6-HT4	+0.07628	-3.917	Fail physical
G1.6-H2-NOVT	+0.05046	-0.284	Fail
G1.6-H2-LONG	+0.05441	-0.278	Rejected; campaign closed

The analysis-only fairing extension to FS 11.500 m is not the production 10.750 m fuselage. G1.4–G1.6 script labels identify diagnostic revisions. The accepted production clean model remains the thin lifting system plus separate PDT parasite drag; the fuselage stability contribution is treated analytically in Section 6.

5. Stability and control-derivative evidence

5.1 Baseline continuity and sideslip response

Table 34. G5 baseline coefficient continuity — panel A

Test	C_L	$C_{L,w}$	C_{D_i}	$C_{D_i,w}$
CA-G3 A4 reference	0.825049	0.826879	0.0302665	0.0226181
G5 B0.1 A4	0.8250488	0.8268787	0.0302665	0.0226181
G5 B0.2 A0/B0	0.4338828	0.4347331	0.0089078	0.0063150

Table 35. G5 baseline coefficient continuity — panel B

Test	C_{M_y}	C_S	C_{M_x}	C_{M_z}
CA-G3 A4 reference	+0.002280	+0.000286	−0.000203	+0.0000933
G5 B0.1 A4	+0.0022801	+0.0002864	−0.0002031	+0.0000933
G5 B0.2 A0/B0	+0.0949899	+0.0012737	−0.0008886	+0.0004578

Table 36. G5 baseline iteration and residual checks — panel A

Test	$ \Delta CL $	$ \Delta CL,w $	$ \Delta CD_i,w $	$ \Delta CMy $
B0.1 A4	0.208%	0.220%	0.272%	0.003026
B0.2 A0/B0	0.012%	0.018%	0.171%	6.0×10^{-7}

Table 37. G5 baseline iteration and residual checks — panel B

Test	log10L2	log10Rmax	Status
B0.1 A4	−0.866	+0.631	PASS
B0.2 A0/B0	−1.756	−0.256	PASS

B0.1 reproduces CA-G3 at essentially numerical precision. The zero-angle B0.2 anchor has a small side-force intercept of +0.001274. That intercept is retained in fits rather than forced to zero; conventional roll and yaw signs follow Section 1.4. The accepted manual sideslip sweep covers -6° to $+6^\circ$, with the $\pm 4^\circ$ fit primary.

Table 38. G5 manual sideslip sweep — panel A

Test	β	C_Y	C_l	C_n
B1.1	-6°	+0.046211	+0.008382	−0.014461
B1.2	-4°	+0.031054	+0.005879	−0.009733
B1.3	-2°	+0.016085	+0.003389	−0.005064
B1.4	0°	+0.001274	+0.000887	−0.000458

Test	β	C_Y	C_I	C_n
B1.5	+2°	-0.013564	-0.001630	+0.004156
B1.6	+4°	-0.028411	-0.004143	+0.008777
B1.7	+6°	-0.043394	-0.006657	+0.013440

Table 39. G5 manual sideslip sweep — panel B

Test	C_L	C_m	$C_{D_{i,w}}$	Status
B1.1	0.432255	+0.079014	0.008134	PASS
B1.2	0.433493	+0.086431	0.007166	PASS
B1.3	0.434142	+0.091409	0.006565	PASS
B1.4	0.433879	+0.095006	0.006315	PASS
B1.5	0.432825	+0.096526	0.006433	PASS
B1.6	0.430968	+0.096246	0.006917	PASS
B1.7	0.428320	+0.094423	0.007779	PASS

Table 40. Static lateral-directional derivative fits

Quantity	Primary $\pm 4^\circ$ linear fit	R^2	All-seven fit	Assessment
$C_{Y\beta}$	-0.42565 rad ⁻¹	0.999997	-0.42705	PASS
$C_{I\beta}$	-0.07180 rad ⁻¹	0.999995	-0.07180	PASS
$C_{n\beta}$	+0.13247 rad ⁻¹	0.999995	+0.13295	PASS

Table 41. Sideslip-sweep validation gates

Gate	Result	Status
B0.2 zero-beta repeatability	C_L difference 0.00093%	PASS
Central-range linearity	all $R^2 > 0.99999$	PASS
$\pm 2^\circ$ symmetry	max error 0.59%	PASS
$\pm 4^\circ$ symmetry	max error 0.76%	PASS
$\pm 6^\circ$ symmetry	max error 0.75%	PASS
Central-difference derivative consistency	<1% variation	PASS
Final-step C_L convergence	all <0.05%	PASS

Gate	Result	Status
Final-step $C_{D_i,w}$ convergence	all <0.20%	PASS
Directional static stability	$C_{n\beta} = +0.13247$	PASS, lifting-system basis
Dihedral effect	$C_{l\beta} = -0.07180$	PASS
Overall 01_Beta_Sweep	—	PASS / CLOSED

5.2 Steady derivatives and direct damping

Table 42. Manual and built-in steady-derivative comparison

Quantity	Manual/ reference	VSPAERO Steady	Difference	Disposition
$C_{Y\beta}$	-0.42565	-0.41767	1.88%	PASS
$C_{l\beta}$	-0.07180	-0.08204	14.26%	SENSITIVITY — retain manual
$C_{n\beta}$	+0.13247	+0.13007	1.81%	PASS
$C_{L\alpha}$	+5.62514 ($\pm 2^\circ$)	+5.52430	1.79%	PASS
$C_{m\alpha}$	-1.17352 ($\pm 2^\circ$)	-1.02998	12.23%	SENSITIVITY — retain finite- difference fit
X_{NP}	4.1424 m ($\pm 2^\circ$)	4.1007 m	-0.0417 m	Bound for Section 6

Table 43. Executed direct damping and cross-rate derivatives

Derivative	Value	Physical interpretation	Status
C_{lp}	-0.55079	roll damping	Stabilising
C_{mq}	-19.25475	pitch damping	Stabilising
C_{nr}	-0.11230	yaw damping	Stabilising
C_{np}	-0.02398	roll–yaw coupling	Retain
C_{lr}	+0.12822	yaw–roll coupling	Retain
C_{Yp}	-0.12061	roll-sideforce coupling	Retain
C_{Yr}	+0.31553	yaw-sideforce coupling	Retain

The manual multi-point static derivatives govern where the built-in steady estimate differs. In particular, the source reports 14.3% difference in dihedral derivative and 12.23% difference in

pitch-stiffness derivative. Negative roll, pitch and yaw direct-damping derivatives support bounded stability screens; they do not establish Dutch-roll or spiral eigenvalues without a controlled inertia tensor and coupled state matrix.

5.3 Elevator, aileron and rudder effectiveness

Table 44. Control-group definition

Group	Actual surfaces	Role
0 MainWing_SS_CONT_0	two MainWing copies	Aileron or flap — identify later
1 MainWing_SS_CONT_1	two MainWing copies	Flap or aileron — identify later
2 HTail_SS_CONT_0	HTail_Surf0_SS_CONT_0 + HTail_Surf1_SS_CONT_0	Elevator
3 VTail_SS_CONT_0	VTail control surface	Rudder

Positive elevator command is trailing-edge down. Positive aileron command is starboard trailing-edge down and port trailing-edge up. Positive rudder command is trailing-edge right. The coefficient signs below are the conventional mapped signs, not unmodified VSPAERO moment-axis labels.

Table 45. Elevator run results — panel A

Test ID	δ_e	C_L	$C_{L,w}$
E3.1	−5° TE-up	0.365765	0.367051
E3.0	0°	0.434055	0.434971
E3.2	+5° TE-down	0.502450	0.503002

Table 46. Elevator run results — panel B

Test ID	$C_{D_i,w}$	C_m	Status
E3.1	0.006435	+0.292504	PASS
E3.0	0.006324	+0.094596	PASS
E3.2	0.008488	−0.102068	PASS

Table 47. Elevator-effectiveness estimates

Quantity	Negative-side estimate	Positive-side estimate	Central estimate	Linearity
$C_{m\delta_e}$	−2.26786	−2.25360	−2.26073 rad ^{−1}	0.63% mismatch
$C_{L\delta_e}$	+0.78255	+0.78375	+0.78315 rad ^{−1}	0.15% mismatch

The accepted central elevator derivatives are pitch moment −2.26073 rad^{−1} and lift +0.78315 rad^{−1}. The small-deflection derivative is not extrapolated as a validated stall or separated-flow model.

Table 48. Aileron run results — panel A

Test ID	δa	C_Y	C_l	C_n
A4.1	-5°	+0.000661	+0.007990	-0.000888
A4.0	0°	+0.001267	+0.000882	-0.000454
A4.2	$+5^\circ$	+0.001859	-0.006216	-0.000019

Table 49. Aileron run results — panel B

Test ID	C_L	C_m	$C_{D_{l,w}}$	Status
A4.1	0.434049	+0.094826	0.006597	PASS
A4.0	0.434055	+0.094596	0.006324	PASS
A4.2	0.434041	+0.094706	0.006605	PASS

Table 50. Aileron-effectiveness estimates

Derivative	$-5 \rightarrow 0$ estimate	$0 \rightarrow +5$ estimate	Central estimate	One-sided mismatch
$C_{l\delta_a}$	-0.081445	-0.081341	-0.081393 rad^{-1}	0.13%
$C_{n\delta_a}$	+0.004971	+0.004987	+0.004979 rad^{-1}	0.31%
$C_{Y\delta_a}$	+0.006954	+0.006782	+0.006868 rad^{-1}	2.51%

Table 51. Rudder run results — panel A

Test	δr	C_Y	C_l	C_n
R5.1	-5°	+0.027319	+0.003236	-0.010117
R5.0	0°	+0.001267	+0.000882	-0.000454
R5.2	$+5^\circ$	-0.024697	-0.001538	+0.009179

Table 52. Rudder run results — panel B

Test	C_L	C_m	$C_{D_{l,w}}$	Status
R5.1	0.433364	+0.097559	0.006840	PASS
R5.0	0.434048	+0.094831	0.006324	PASS
R5.2	0.434109	+0.095377	0.007102	PASS

Table 53. Rudder-effectiveness estimates

Derivative	-5→0	0→+5	Central	One-sided mismatch
$C_{n\delta_r}$	+0.11073	+0.11039	+0.11056 rad ⁻¹	0.31%
$C_{Y\delta_r}$	-0.29852	-0.29753	-0.29803 rad ⁻¹	0.33%
$C_{l\delta_r}$	-0.02697	-0.02773	-0.02735 rad ⁻¹	2.78%

Table 54. Control-sweep interpretation

Control	Accepted central derivatives (rad ⁻¹)	Use / limitation
Elevator	$C_{m\delta_e} = -2.26073$; $C_{L\delta_e} = +0.78315$	Small-deflection longitudinal control baseline
Aileron	$C_{l\delta_a} = -0.081393$; $C_{n\delta_a} = +0.0049793$; $C_{Y\delta_a} = +0.006868$	Linear lateral balance; VMC applies a 25% effectiveness debit
Rudder	$C_{n\delta_r} = +0.11056$; $C_{Y\delta_r} = -0.29803$; $C_{l\delta_r} = -0.02735$	Linear directional balance; VMC applies a 15% effectiveness debit

5.4 Local longitudinal stability and reference invariance

Table 55. SM6.1 local angle-of-attack sweep — panel A

Test	α	C_L	$C_{L,w}$	C_m
SM6.1a	-2°	0.237455	0.238195	+0.133491
SM6.1b	-1°	0.335829	0.336586	+0.114606
SM6.1c	0°	0.434050	0.434966	+0.094832
SM6.1d	+1°	0.531665	0.532832	+0.073838
SM6.1e	+2°	0.629870	0.631211	+0.051466

Table 56. SM6.1 local angle-of-attack sweep — panel B

Test	$C_{D_{i,w}}$	19→20 CL	19→20 Cm	Status
SM6.1a	0.002460	0.0152%	0.000109	PASS
SM6.1b	0.004043	0.0085%	0.000030	PASS
SM6.1c	0.006324	0.0094%	0.000039	PASS
SM6.1d	0.009349	0.0480%	0.000011	PASS
SM6.1e	0.013052	0.0312%	0.000155	PASS

Table 57. SM6.1 lifting-system derivative fit

Quantity	Result	Assessment
C_{L0}	0.433774	Accepted
$C_{L\alpha}$	+5.61880 rad ⁻¹	PASS
$R^2(C_L)$	0.9999984	Excellent
C_{m0}	+0.093647	Accepted
$C_{m\alpha}$	-1.17352 rad ⁻¹	Statically stable
$R^2(C_m)$	0.998852	Slight curvature; acceptable
$\partial C_m / \partial C_L$	-0.20885	Stable
$x_{NP,LS}$	4.14286 m	Accepted
$h_{N,LS}$	58.10% MAC	Lifting system only

The uncorrected lifting-system neutral point is FS 4.14286 m. It is not a full-aircraft neutral point: the destabilising fuselage term has not yet been added. The initial statement that the optional reference audit could be skipped is superseded by the subsequent executed SM6.2 test.

Table 58. Executed SM6.2 moment-reference audit

Gate	Result	Status
x_{ref}	3.900 m	Confirmed
Alpha points	-2°, 0°, +2°	Confirmed
C_L repeatability vs SM6.1	Numerically identical	PASS
Moment translation	max error 5.0×10^{-5}	PASS
$C_{L\alpha}$	+5.62093 rad ⁻¹	PASS
$C_{m\alpha}$	-0.72528 rad ⁻¹	PASS
x_{NP}	4.14271 m	PASS
Same-method NP difference	0.000467 m	PASS
NP difference as MAC	0.0248%	PASS
20 iterations	all three	PASS
Force/Trefftz convergence	all gates satisfied	PASS
Overall SM6.2	—	PASS / CLOSED

Moving the reference from FS 3.750 to 3.900 m preserves lift and induced drag and reproduces moment translation within 5.0×10^{-5} . The matched three-point neutral points differ by only 0.47 mm. The lifting-system result is therefore reference-invariant to the recorded numerical precision.

6. Fuselage correction and CG screening

6.1 Method and controlled geometry

SM6.3B applies the recorded Gudmundsson Appendix C1 Munk–Multhopp implementation to the G1.3 fuselage. Body width, local flow-angle gradient and segment length determine the destabilising moment slope. The 36.5 coefficient produces a per-degree slope; conversion to per radian is explicit. All dimensions are used consistently in metres.

Table 59. SM6.3A isolated MainWing support sweep

α	C_L	$C_{L,w}$	$C_{D_{l,w}}$	C_m
-2°	0.315070	0.315273	0.003283	-0.068048
0°	0.485915	0.486460	0.007778	-0.046735
$+2^\circ$	0.656363	0.657540	0.014217	-0.030239

The isolated-wing lift-curve slope is 4.88866 rad^{-1} and its aerodynamic centre is approximately FS 3.5416 m (26.1% MAC). Comparison with the analytical wing-only value 4.837 rad^{-1} differs by +1.07%. The complete lifting-system slope 5.61880 rad^{-1} is not the appropriate wing-only comparison.

Table 60. SM6.3B controlled inputs

Controlled quantity	Value	Evidence
Wing reference area, S	30.000 m ²	G5 / SM6.1
Reference chord, $\bar{c}$	1.881 m	G5 / SM6.1
Wing span, b	16.400 m	G1.3
Aspect ratio	8.96533	Derived
Moment reference	FS 3.750 m	SM6.1
MAC leading edge	FS 3.050 m	G1.3
Wing root LE / TE	FS 2.930079 / 5.290425 m	G1.3
HT aerodynamic centre	FS 9.420166 m	G1.3
TE-to-HT-AC distance, l_h	4.129741 m	Derived
MainWing-only $C_{L\alpha,w}$	4.88866 rad^{-1}	SM6.3A
Lifting-system $C_{L\alpha}$	5.61880 rad^{-1}	SM6.1
Lifting-system $C_{m\alpha}$	$-1.17352 \text{ rad}^{-1}$	SM6.1
Lifting-system neutral point	FS 4.14286 m	SM6.1 / SM6.2

$$C_{m\alpha, FUS, deg} = \frac{1}{36.5 S \bar{c}} \sum_i w_{f,i}^2 \left(\frac{\partial \beta}{\partial \alpha} \right)_i \Delta x_i \quad (6)$$

$$C_{m\alpha, FUS, rad} = \frac{180}{\pi} C_{m\alpha, FUS, deg} \quad (7)$$

$$AR = \frac{b^2}{S} = 8.96533, \frac{d\epsilon}{d\alpha} = \frac{2C_{L\alpha, w}}{\pi AR} = 0.347139 \quad (8)$$

For forward-body stations, x is the positive distance from the wing-root leading edge to the segment midpoint. The branch expressions below retain the source implementation; the wing-covered interval contributes zero.

$$\left(\frac{\partial \beta}{\partial \alpha}\right)_1 = 1.4758 - 0.40485 \frac{x}{\bar{c}} + 0.09107 \left(\frac{x}{\bar{c}}\right)^2, x/\bar{c} \geq 1 \quad (9)$$

$$\left(\frac{\partial \beta}{\partial \alpha}\right)_2 = 1.74611 \left(\frac{x}{\bar{c}}\right)^{-0.4254}, x/\bar{c} < 1 \quad (10)$$

For the aft body, x_i is the distance aft of the wing-root trailing edge and l_h is the root-trailing-edge to horizontal-tail aerodynamic-centre distance.

$$\left(\frac{\partial \beta}{\partial \alpha}\right)_i = \frac{x_i}{l_h} \left(1 - \frac{d\epsilon}{d\alpha}\right) \quad (11)$$

Table 61. Frozen G1.3 fuselage stations

G1.3 station	FS x (m)	Width (m)	Height (m)
Nose point	0.0000	0.000	0.000
Nose	0.5375	0.800	0.900
Cabin front	2.3650	1.550	1.700
Cabin rear	5.3750	1.550	1.700
Upsweep	7.0950	1.250	1.340
Tail boom	9.1375	0.780	0.800
Tail point	10.7500	0.000	0.000

Widths are linearly interpolated; each strip uses the mean of its endpoint widths. Height is retained for geometry traceability but is not an input to this width-based slope equation. The root-wing-covered interval is FS 2.930079–5.290425 m.

6.2 Station-aligned integration and central result

Table 62. S14 forward and wing-covered strip integration — panel A

ID	x_0 - x_1 (m)	Δx	w_f
F1	0.0000-0.5375	0.5375	0.4000
F2	0.5375-1.1467	0.6092	0.9250
F3	1.1467-1.7558	0.6091	1.1750
F4	1.7558-2.3650	0.6092	1.4250

ID	x_0-x_1 (m)	Δx	w_f
F5	2.3650-2.9301	0.5651	1.5500
W	2.9301-5.2904	2.3603	1.5500

Table 63. S14 forward and wing-covered strip integration — panel B

ID	$x/\bar{c}$ or x/l_h	$\partial\beta/\partial\alpha$	Term (m ³)
F1	1.4148	1.0853	0.09334
F2	1.1100	1.1386	0.59351
F3	0.7862	1.9343	1.62659
F4	0.4623	2.4243	2.99902
F5	0.1502	3.9112	5.30982
W	-	0.0000	0.00000

Table 64. S14 aft-body strip integration — panel A

ID	x_0-x_1 (m)	Δx	w_f
A1	5.2904-5.3750	0.0846	1.5500
A2	5.3750-6.2350	0.8600	1.4750
A3	6.2350-7.0950	0.8600	1.3250
A4	7.0950-8.1163	1.0213	1.1325
A5	8.1163-9.1375	1.0212	0.8975
A6	9.1375-9.6750	0.5375	0.6500
A7	9.6750-10.2125	0.5375	0.3900
A8	10.2125-10.7500	0.5375	0.1300

Table 65. S14 aft-body strip integration — panel B

ID	$x/\bar{c}$ or x/l_h	$\partial\beta/\partial\alpha$	Term (m ³)
A1	0.0102	0.0067	0.00136
A2	0.1246	0.0813	0.15221
A3	0.3328	0.2173	0.32809
A4	0.5606	0.3660	0.47942
A5	0.8079	0.5275	0.43387
A6	0.9966	0.6507	0.14776

ID	$x/\bar{c}$ or x/l_h	$\partial \beta / \partial \alpha$	Term (m ³)
A7	1.1268	0.7356	0.06014
A8	1.2569	0.8206	0.00745

The source's independent station audit records 10.622273 m³ forward and 1.610303 m³ aft, giving 12.232575 m³ total. The forward body contributes 86.8% of the destabilising integral. The following equations express the recorded central result with native equation formatting.

$$C_{m\alpha, FUS, deg} = \frac{12.232575}{36.5 (30.000) (1.881)} = 0.005939 \text{ deg}^{-1} \quad (12)$$

$$C_{m\alpha, FUS} = +0.34028 \text{ rad}^{-1} \quad (13)$$

$$C_{m\alpha, corr} = -1.17352 + 0.34028 = -0.83324 \text{ rad}^{-1} \quad (14)$$

$$x_{NP, corr} = x_{ref} - \dot{c} \frac{C_{m\alpha, corr}}{C_{L\alpha, LS}} = 4.02894 \text{ m} \quad (15)$$

$$SM(x_{CG}) = \frac{x_{NP, corr} - x_{CG}}{\dot{c}} \quad (16)$$

Table 66. Central corrected neutral point and former design-boundary checks

Item	FS x (m)	% MAC	Assessment
Lifting-system NP	4.14286	58.10%	Reference-invariant thin model
Fuselage-corrected NP	4.02894	52.04%	Central conceptual estimate
Forward design CG	3.60000	29.24%	SM = 22.80%
Former aft design CG	3.90000	45.19%	SM = 6.86% - FAIL
NP required for 10% at FS 3.900	4.08810	55.19%	Required
Central aft CG for exactly 10%	3.84084	42.04%	Provisional central limit

FS 3.900 m yields only 6.86% MAC margin on the central model and fails the 10% floor. It is not retained as an acceptable aft operating boundary. This is a fuselage-moment-corrected lifting-system estimate; it omits nacelle terms, powered-propeller normal-force derivatives and a separate fuselage lift-curve contribution.

6.3 Reproducible discretisation and model-form limits

Table 67. Defined strip and quadrature sensitivities — panel A

Case	Definition	Regions	$C_{m\alpha, FUS}$
U11	4 F + 1 W + 6 A, uniform by region	11	0.32977

Case	Definition	Regions	$C_{m\alpha, FUS}$
U14	5 F + 1 W + 8 A, uniform by region	14	0.33711
S14	5 F + 1 W + 8 A, station-aligned	14	0.34028
U20	8 F + 1 W + 11 A, uniform by region	20	0.34943
Quadrature check	Fine continuous-width midpoint audit; not selected	-	0.38902

Table 68. Defined strip and quadrature sensitivities — panel B

Case	x_{NP}	SM @ 3.90	10% aft limit
U11	4.03246	7.04%	3.84436
U14	4.03000	6.91%	3.84190
S14	4.02894	6.86%	3.84084
U20	4.02588	6.69%	3.83778
Quadrature check	4.01263	5.99%	3.82453

U11, U14 and U20 use uniform subdivisions within the forward and aft regions; S14 is station-aligned. All defined meshes fail the former FS 3.900 m boundary. U20 is the most restrictive defined 10-to-20-region screen and sets FS 3.83778 m. The finer midpoint quadrature is a warning about the integrable near-leading-edge upwash singularity, not a substituted release method. Agreement between strip meshes does not validate the classical model for this high-wing, rounded-rectangle fuselage.

$$x_{CG,max,U20} = 4.02588 - 0.10(1.881) = 3.83778 \text{ m} \quad (17)$$

6.4 Loading-cloud evidence and release boundary

$$x_{CG,k} = \frac{\sum_j m_{j,k} x_j}{\sum_j m_{j,k}}, h_{CG,k} = \frac{x_{CG,k} - 3.050}{1.881} \quad (18)$$

$$SM_{rob,k} = \frac{x_{NP,corr} - \Delta x_{NP,model} - x_{CG,k} - \Delta x_{CG,k}}{\dot{c}} \geq 0.10 \quad (19)$$

The early SM6.4 verification issue could not construct a loading cloud because component arms were absent from that issue. Later records identify nominal loading cases and an integrated CG closure document, and those cases are used by VMC and FG8. This compendium preserves that progression without claiming the underlying component-arm ledger was supplied or independently checked for this editorial issue.

Table 69. Later source-reported CG uncertainty results

Case	Reported CG station	Disposition
Nominal loading cloud	FS 3.34618–3.70312 m	Inside the U20 aft screen; forward control checked separately in FG8
Independent 95% aft sensitivity	FS 3.76080 m	Below FS 3.83778 m; source summary reports approximately 14.09% margin
Deterministic all-at-bounds aft case	FS 3.91324 m	Beyond the U20 aft screen; HOLD, not an operational loading case
U20 screening boundary	FS 3.83778 m	10% MAC on the defined U20 model; not a certification-approved envelope

Provenance qualification. The source pairs rounded CG and static-margin summaries that are not all exactly reproduced by the same rounded neutral point. The containment decisions above use the explicit CG stations and U20 boundary; no additional precision is asserted. The integrated ledger, uncertainty convention and the FDR's later 2.0 kg tank-system increment must be reconciled at their recorded arms before operational loading release.

Table 70. Minimum evidence for operational CG release

Ledger block	Required record
Empty aircraft	Component masses, datum-referenced longitudinal arms and source/CAD centroid evidence; growth and credits explicitly located
Occupants / cargo	Each seat and cargo-zone mass/arm; occupancy permutations and loading constraints
Fuel	Tank and unusable-fuel arms through the fuel-burn range; later tank-system mass increment at its controlled arm
Acceptance	All dispatch/arrival and loading-sequence states checked against forward control, aft stability, MTOM and MLM; uncertainty method identified

7. Powered-solver diagnostics and rejection evidence

7.1 VMC7.1 unpowered low-speed baseline

Table 71. VMC7.1 baseline settings

Setting	Value
Alpha / beta	0 / 0 degrees
Fowler, aileron, elevator, rudder	0 / 0 / 0 / 0 degrees
Actuator disks	Off / zero loading
V_{∞}	20.58 m/s
Mach / Reynolds number	0.0605 / 2.65×10^6
Density	1.225 kg/m ³
Reference station	FS 3.70312 m
Wake iterations	20

Table 72. VMC7.1 final coefficients and final-five spans

Quantity	Iteration-20 value	Final-five span	Gate	Result
C_L	0.427911	0.0898%	less than 0.5%	PASS
$C_{L,w}$	0.428777	0.0656%	less than 0.5%	PASS
$C_{D_i,w}$	0.006182	0.201%	less than 0.5%	PASS
C_m	+0.083816	0.000130	less than 0.005	PASS
C_Y	+0.001239	0.000204	less than 5×10^{-4}	PASS
C_l	+0.000870	0.000061	less than 5×10^{-4}	PASS
C_n	-0.000447	0.000015	less than 5×10^{-4}	PASS

VMC7.1 establishes an unpowered low-speed numerical anchor. VMC7.2 then checks the actual flap-group implementation. These are configuration-verification tests; marginal wake-drag spans at high angle are not promoted as production performance evidence.

7.2 VMC7.2 Fowler grouping and corrected sweep

Table 73. Rejected initial flap grouping versus corrected configuration

Alpha = 0 degrees quantity	Initial file	Corrected file	Change
C_L	0.427900	0.846943	Fowler lift increment recovered
C_Y	+0.011656	+0.001657	85.8% reduction in magnitude
C_l	-0.063141	+0.001371	97.8% reduction in magnitude
C_n	-0.002730	-0.000641	76.5% reduction in magnitude
C_m	+0.078101	+0.174848	Corrected flap load/moment state

The initial configuration produced conventional roll coefficient -0.063141 at zero angle; the corrected grouping gives $+0.001371$. The sign and magnitude change identify a configuration error, not a physical aircraft asymmetry. The corrected symmetric $+12^\circ$ Fowler state is retained.

Table 74. Corrected $+12^\circ$ Fowler longitudinal sweep

Alpha (degrees)	C_L	$C_{L,w}$	$C_{D_i,w}$	C_m
0	0.846943	0.851300	0.027738	+0.174848
4	1.228337	1.234087	0.052715	+0.061235
8	1.604105	1.614415	0.087709	-0.072316

Table 75. Corrected $+12^\circ$ Fowler lateral checks

Alpha (degrees)	C_Y	C_l	C_n	Lateral-state interpretation
0	+0.001657	+0.001371	-0.000641	Small corrected residual
4	+0.000492	+0.000394	-0.000175	Approaching zero
8	-0.000689	-0.000443	+0.000189	Gentle sign reversal

Table 76. Corrected Fowler final-five convergence audit

Alpha (degrees)	C_L span	$C_{L,w}$ span	$C_{D_i,w}$ span	C_m absolute span	Assessment
0	0.070%	0.069%	0.105%	0.000265	PASS

Alpha (degrees)	C_L span	$C_{L,w}$ span	$C_{D,w}$ span	C_m absolute span	Assessment
4	0.639%	0.612%	0.750%	0.001931	Minor wake-span exceedance; trend retained
8	0.123%	0.175%	0.805%	0.000439	Minor wake-drag span exceedance; trend retained

7.3 Actuator-disk input physics

$$J = \frac{V}{nD}, T = C_T \rho n^2 D^4, P = C_P \rho n^3 D^5 \quad (20)$$

$$\eta_p = \frac{TV}{P} = J \frac{C_T}{C_P} \quad (21)$$

The initial pair $C_t = 0.40$ and $C_p = 0.07454$ gives efficiency 1.667 and is rejected as physically impossible. With $C_t = 0.1490$, $C_p = 0.07454$, 2.0 m diameter, 2,279 rpm, 23.60 m/s and density 1.225 kg/m³, the diagnostic input corresponds to 4.213 kN thrust, 160.12 kW and efficiency approximately 0.621. Correcting the disk inputs is necessary but does not establish convergence of the coupled powered model.

Table 77. Early powered diagnostic branches

Source	Alpha = 0 degrees final state	Diagnostic conclusion
Initial VMC7_3...csv	$C_L = 0.88396$, $C_Y = +0.00725$, $C_l = -0.09143$, $C_m = +0.49504$	Dominated by flap-grouping error; reject
Corrected-grouping VMC7_3...(1).csv	$C_L = 1.26838$, $C_Y = -0.00704$, $C_l = +0.01540$, $C_m = +0.64453$	Flap asymmetry removed, but powered iteration remains oscillatory and old C_T basis is invalid

7.4 Controlled C–F comparison

Table 78. Powered diagnostic settings

Setting	VMC7.3C	VMC7.3D	VMC7.3E	VMC7.3F
Alpha / beta	0 / 0 degrees	0 / 0 degrees	0 / 0 degrees	0 / 0 degrees
Fowler / other controls	+12 / 0 degrees	+12 / 0 degrees	+12 / 0 degrees	+12 / 0 degrees
V_∞	23.60 m/s	23.60 m/s	23.60 m/s	23.60 m/s
RPM magnitude	2279	2279	2279	2279

Setting	VMC7.3C	VMC7.3D	VMC7.3E	VMC7.3F
C_T / C_P	0.1490 / 0.07454	Same	Same	Same
Rotation architecture	Co-rotating	Co-rotating	Counter-rotating	Counter-rotating
Wake nodes / iterations	16 / 40	16 / 40	16 / 40	16 / 40
Fixed wake	Off	Off	Off	Off
Relaxation	1.0	0.5	0.5	0.25

Table 79. Powered final longitudinal coefficients

Test	C_L	$C_{L,w}$	$C_{D_{l,w}}$	C_m
VMC7.3C	1.27814	1.41405	0.031024	+0.62514
VMC7.3D	1.28147	1.41374	0.032805	+0.61839
VMC7.3E	1.29625	1.45414	0.043366	+0.66299
VMC7.3F	1.34860	1.48845	0.045933	+0.49502

Table 80. Powered final lateral coefficients

Test	$ C_Y $	$ C_I $	$ C_n $	Lateral interpretation
VMC7.3C	0.010801	0.016040	0.005146	Co-rotating asymmetry
VMC7.3D	0.007525	0.016238	0.003925	Asymmetry persists
VMC7.3E	0.001742	0.002991	0.000582	Counter-rotation improvement
VMC7.3F	0.001706	0.001993	0.000592	Small instant, still oscillatory

Table 81. Powered final-five longitudinal spans

Test	C_L span	$C_{L,w}$ span	$C_{D_{l,w}}$ span	C_m span
VMC7.3C	3.26%	2.80%	12.48%	0.12857
VMC7.3D	1.77%	0.90%	4.42%	0.07542
VMC7.3E	4.11%	2.11%	3.82%	0.17326
VMC7.3F	3.91%	2.72%	8.24%	0.15738

Table 82. Powered final-five lateral spans

Test	C_Y span	C_l span	C_n span	Lateral gate result
VMC7.3C	0.022017	0.001835	0.008318	FAIL
VMC7.3D	0.004114	0.002461	0.001707	FAIL
VMC7.3E	0.005732	0.000890	0.002242	FAIL
VMC7.3F	0.005705	0.000882	0.002167	FAIL

Table 83. Powered residual audit

Test	Final $\log_{10} L_2$	Final $\log_{10} R_{max}$	Observation
VMC7.3C	-0.2859	+1.0153	Oscillatory, no sustained decay
VMC7.3D	-0.3463	+1.3910	Relaxation did not close residuals
VMC7.3E	-0.0873	+1.3677	Counter-rotation fixed symmetry, not iteration stability
VMC7.3F	-0.4550	+1.2570	Final numerical test still oscillatory

Table 84. VMC7.3F late-iteration longitudinal coefficients

Iteration	C_L	$C_{L,w}$	$C_{D,w}$	C_m
36	1.29886	1.45337	0.042352	0.65096
37	1.29735	1.44869	0.042604	0.65240
38	1.30070	1.45718	0.042745	0.64391
39	1.30631	1.45525	0.043748	0.62739
40	1.34860	1.48845	0.045933	0.49502

Table 85. VMC7.3F late-iteration lateral coefficients and residuals

Iteration	C_Y	C_l	C_n	$\log_{10} L_2$	$\log_{10} R_{max}$
36	+0.002278	+0.002351	-0.000877	-0.206	1.421
37	+0.002343	+0.002875	-0.000964	-0.354	1.142
38	-0.003362	+0.002782	+0.001203	-0.115	1.541
39	-0.002219	+0.002669	+0.000822	-0.315	1.552
40	+0.001706	+0.001993	-0.000592	-0.455	1.257

7.5 Decision and permitted use

Table 86. Counter-rotation sensitivity at relaxation 0.5

Quantity	Co-rotating RLX 0.5	Counter-rotating RLX 0.5	Magnitude reduction
$ C_Y $	0.007525	0.001742	76.8%
$ C_l $	0.016238	0.002991	81.6%
$ C_n $	0.003925	0.000582	85.2%

Counter-rotation substantially reduces equal-power lateral asymmetry: the recorded magnitude reductions are approximately 76.8%, 81.6% and 85.2% for the compared lateral quantities. This is a useful architecture sensitivity, not a converged powered aerodynamic database. Cases C–F still fail the adopted production convergence gates, including at relaxation 0.25.

Decision. The coupled actuator-disk/free-wake branch is rejected for production coefficients. Neither strongly oscillating final coefficients nor a favourable counter-rotation trend is used to claim powered high-lift, OEI rolling moment or control closure. VMC7.4 instead retains the converged G5 airframe derivatives and introduces explicitly bounded external propulsion forces and moments.

8. Atmosphere-consistent OEI control equilibrium

8.1 Governing basis and loading cases

VMC7.4 is a deterministic steady one-engine-inoperative force-and-moment balance, not a replacement claim of high-fidelity powered CFD. It uses G5 derivatives, CG-specific yaw transfer, bounded live thrust and failed-propeller drag, adverse reaction torque and finite conventional controls. No differential-thrust relief is credited. Left/right engine loss is treated by symmetry with adverse torque sense retained.

Table 87. Recorded VMC loading cases

Case	Mass	CG station	Role
VMC-A	1894.8 kg	FS 3.70312 m	Most-aft actual departure case; primary
VMC-B	2000.0 kg	FS 3.65023 m	Most-aft MTOM case
VMC-C	1900.0 kg	FS 3.63687 m	Most-aft planned landing-mass case

Table 88. G5 derivatives and VMC effectiveness application

Derivative	G5 value (rad ⁻¹)	Applied VMC7.4 value
$C_{Y\beta}$	-0.42565	-0.42565
$C_{l\beta}$	-0.071802	-0.071802
$C_{n\beta}$	+0.13247	CG-transferred
$C_{Y\delta_a}$	+0.006868	0.75 times derivative
$C_{l\delta_a}$	-0.081393	0.75 times derivative
$C_{n\delta_a}$	+0.0049793	CG-transferred, then times 0.75
$C_{Y\delta_r}$	-0.29803	0.85 times derivative
$C_{l\delta_r}$	-0.02735	0.85 times derivative
$C_{n\delta_r}$	+0.11056	CG-transferred, then times 0.85

Table 89. Atmosphere-consistent VMC input register

Input	Controlled value / rule
Reference density for EAS	$\rho_0 = 1.225 \text{ kg/m}^3$
Release atmosphere	Sea-level standard, $\rho = 1.225 \text{ kg/m}^3$
Sensitivity atmosphere	610 m ISA+20°C, $\rho = 1.079 \text{ kg/m}^3$

Input	Controlled value / rule
Wing / span	S = 30.0 m ² ; b = 16.4 m
Propeller diameter / arm	2.00 m / 1.60 m
Live shaft-power bound	160.13 kW at 2279 RPM
Failed-propeller drag states	CD,f = 0.08, 0.25, 0.45, 0.60
Control effectiveness debit	75% aileron; 85% rudder
Physical travel / sideslip / bank	±22° aileron; ±35° rudder; β ≤5°; φ ≤5°
Internal VMC gate	VMC ≤ 0.95 VR = 39.96 kn EAS

These loading cases are source-recorded analysis states. They are not silently reconstituted using the FDR's later fuel or tank-system mass. Confirmation at the updated component-arm ledger remains a configuration-control requirement.

$$C_{n\chi}(x_{CG}) = C_{n\chi}(x_{ref}) + \frac{x_{CG} - x_{ref}}{b} C_{Y\chi}, \chi \in \{\beta, \delta_a, \delta_r\} \quad (22)$$

The control derivatives are debited by $\eta_a = 0.75$ and $\eta_r = 0.85$. In the matrix below, yaw-control entries marked “eff” already include the corresponding debit after reference transfer; they must not be debited twice.

Table 90. CG-transferred yaw derivatives

Case	$C_{n\beta}$	$C_{n\delta_a, eff}$	$C_{n\delta_r, eff}$
VMC-A	0.133687	0.003720	0.094700
VMC-B	0.135059	0.003703	0.095517
VMC-C	0.135406	0.003699	0.095723

8.2 Atmosphere, thrust, failed-propeller drag and torque

$$q = \frac{1}{2} \rho_0 V_E^2, V_{TAS} = V_E \sqrt{\frac{\rho_0}{\rho}} \quad (23)$$

$$A_p = \frac{\pi D_p^2}{4} = 3.1416 \text{ m}^2 \quad (24)$$

$$P_{live} = 2 \rho A_p v_i (V_{TAS} + v_i)^2 \quad (25)$$

$$T_{live} = 2 \rho A_p v_i (V_{TAS} + v_i) \quad (26)$$

$$D_f = q A_p C_{D,f}, C_{n,OEI} = \frac{(T_{live} + D_f) y_p}{q S b} \quad (27)$$

$$\omega = 2 \pi \frac{2279}{60} = 238.66 \text{ rad s}^{-1}, Q_p = \frac{P_{live}}{\omega} \approx 671 \text{ N m} \quad (28)$$

$$C_{l,Q} = s_Q \frac{Q_p}{q S b}, s_Q \in \{-1, +1\} \quad (29)$$

The high-thrust bound uses 160.13 kW live shaft power. Airframe forces use EAS dynamic pressure; ideal momentum thrust uses the matching local TAS and density. Sea-level standard (1.225 kg/m³) governs the reported EAS screen, with 610 m ISA+20°C (1.079 kg/m³) as an operating sensitivity. The ideal-power thrust model is a bound, not a supplier propeller map.

Table 91. Failed-propeller drag envelope

Failed-propeller state	$C_{D,f}$	Use
Feathered / low drag	0.08	Favourable bound
Intermediate / stopped	0.25	Sensitivity
Windmilling control case	0.45	Primary VMC7.4 case
Severe windmilling bound	0.60	Adverse sensitivity

8.3 Simultaneous finite-control solution

$$C_{Y\beta}\beta + \eta_a C_{Y\delta_a}\delta_a + \eta_r C_{Y\delta_r}\delta_r + \frac{W \sin \phi}{q S} = 0 \quad (30)$$

$$C_{l\beta}\beta + \eta_a C_{l\delta_a}\delta_a + \eta_r C_{l\delta_r}\delta_r + C_{l,Q} + C_{l,pw} = 0 \quad (31)$$

$$C_{n\beta,CG}\beta + C_{n\delta_a,CG,eff}\delta_a + C_{n\delta_r,CG,eff}\delta_r + C_{n,OEI} = 0 \quad (32)$$

$$\begin{bmatrix} C_{Y\beta} & \eta_a C_{Y\delta_a} & \eta_r C_{Y\delta_r} \\ C_{l\beta} & \eta_a C_{l\delta_a} & \eta_r C_{l\delta_r} \\ C_{n\beta,CG} & C_{n\delta_a,CG,eff} & C_{n\delta_r,CG,eff} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ \delta_r \end{bmatrix} = - \begin{bmatrix} W \sin \phi / (q S) \\ C_{l,Q} + C_{l,pw} \\ C_{n,OEI} \end{bmatrix} \quad (33)$$

$$|\delta_a| \leq 22^\circ, |\delta_r| \leq 35^\circ, |\beta| \leq 5^\circ, |\phi| \leq 5^\circ \quad (34)$$

$$U(V) = \min_{|\phi| \leq 5^\circ} \max \left(\frac{|\beta|}{5^\circ}, \frac{|\delta_a|}{\delta_{a,lim}}, \frac{|\delta_r|}{\delta_{r,lim}} \right) \quad (35)$$

$$V_{MC} = \min \{ V : U(V) \leq 1 \} \quad (36)$$

At each trial speed, local TAS and density determine thrust; failed-propeller drag and reaction torque determine the external moments. Bank is searched within $\pm 5^\circ$, and the three-equation system is solved for sideslip, aileron and rudder. The minimum feasible speed respects all simultaneous limits. The 90%-travel screen changes the aileron/rudder limits to 19.8° / 31.5° without relaxing bank or sideslip.

8.4 Sea-level-standard release screen

Table 92. Atmosphere-corrected sea-level VMC cases — panel A

Case	CD,f	Failed-prop state	VMC EAS (kn)	Tlive (kN)	Dfail (kN)
VMC-A	0.08	Feathered	38.06	4.472	0.059

Case	CD,f	Failed-prop state	VMC EAS (kn)	Tlive (kN)	Dfail (kN)
VMC-A	0.25	Intermediate	38.47	4.458	0.188
VMC-A	0.45	Windmilling	38.97	4.441	0.348
VMC-A	0.60	Severe	39.37	4.428	0.474
VMC-B	0.08	Feathered	37.92	4.477	0.059
VMC-B	0.25	Intermediate	38.34	4.463	0.187
VMC-B	0.45	Windmilling	38.84	4.446	0.346
VMC-B	0.60	Severe	39.21	4.433	0.470
VMC-C	0.08	Feathered	37.89	4.478	0.058
VMC-C	0.25	Intermediate	38.30	4.464	0.187
VMC-C	0.45	Windmilling	38.80	4.447	0.345
VMC-C	0.60	Severe	39.19	4.434	0.469

Table 93. Atmosphere-corrected sea-level VMC cases — panel B

Case	CD,f	Bank (°)	β (°)	δa (°)	δr (°)
VMC-A	0.08	-3.85	2.72	21.97	35.00
VMC-A	0.25	-3.95	2.81	21.97	35.00
VMC-A	0.45	-4.07	2.97	22.00	34.94
VMC-A	0.60	-4.17	3.07	22.00	34.92
VMC-B	0.08	-3.61	2.69	21.98	35.00
VMC-B	0.25	-3.71	2.77	21.95	35.00
VMC-B	0.45	-3.82	2.96	22.00	34.88
VMC-B	0.60	-3.91	3.02	22.00	34.95
VMC-C	0.08	-3.80	2.68	21.98	35.00
VMC-C	0.25	-3.89	2.77	21.97	35.00
VMC-C	0.45	-4.02	2.95	22.00	34.90
VMC-C	0.60	-4.11	3.03	22.00	34.91

VMC-A governs. The windmilling result is 38.97 kn EAS, 0.99 kn below the 39.96 kn internal gate; severe drag gives 39.37 kn EAS, a 0.59 kn margin. These explicitly supersede the 38.73 / 39.09 kn mixed-density values in Appendix B. “Release screen” identifies the report’s analytical basis, not an approved flight operating limit.

8.5 Hot-high and finite-control sensitivities

Table 94. Hot-high atmosphere sensitivity — panel A

Case	CD,f	State	VMC EAS (kn)	VMC TAS (kn)	Tlive (kN)
VMC-A	0.45	Windmilling	38.28	40.78	4.254
VMC-A	0.60	Severe	38.64	41.17	4.242
VMC-B	0.45	Windmilling	38.14	40.64	4.258
VMC-B	0.60	Severe	38.52	41.04	4.246
VMC-C	0.45	Windmilling	38.09	40.59	4.260
VMC-C	0.60	Severe	38.47	40.99	4.247

Table 95. Hot-high atmosphere sensitivity — panel B

Case	CD,f	Dfail (kN)	δa (°)	δr (°)
VMC-A	0.45	0.336	21.96	35.00
VMC-A	0.60	0.456	22.00	34.99
VMC-B	0.45	0.333	21.95	35.00
VMC-B	0.60	0.453	21.96	35.00
VMC-C	0.45	0.332	21.98	35.00
VMC-C	0.60	0.452	22.00	34.96

Table 96. Control-reserve decision table

Condition	VMC-A result	Gate comparison	Disposition
Full travel, windmilling, sea-level standard	38.97 kn EAS	0.99 kn below 39.96 kn	PASS / THIN
Full travel, severe drag, sea-level standard	39.37 kn EAS	0.59 kn below 39.96 kn	PASS / THIN
90% simultaneous travel, windmilling, sea-level standard	40.97 kn EAS	1.01 kn above 39.96 kn	FAIL internal robustness target
Full travel, windmilling, 610 m ISA+20	38.28 kn EAS	0.70 kn below SL windmill result	Sensitivity only

At 610 m ISA+20°C, VMC-A windmilling is 38.28 kn EAS or 40.78 kn TAS. In the ideal fixed-power thrust model, lower local density reduces thrust at a given EAS, so sea-level standard is the more adverse EAS screen.

At the 39.96 kn gate the corrected windmilling solution requires approximately 20.88° aileron and 33.12° rudder. The additional admissible adverse powered-lift rolling coefficient is only about

0.00206 in the windmill case and 0.00122 in the severe case. These replace the earlier 0.00250 allowance. The 40.97 kn 90%-travel result fails the internal robustness target by 1.01 kn.

Disposition. Conditional analytical PASS at provisional physical stops; robustness FAIL and release HOLD remain visible. Exact control stops, forward/feather/windmill propeller maps, rotation hardware and asymmetric powered-lift roll require higher-fidelity or test evidence. Rejected powered-solver coefficients do not close these items.

9. Forward-CG control and landing-gear evidence

9.1 Scope and recorded loading basis

FG8 addresses forward-CG rotation, landing trim and flare authority, quasi-steady go-around flap transition, and landing-gear reactions. Propulsion remains OFF for the VSPAERO control screens. The +12°, +20° and +35° Fowler states are control-surface surrogates; they do not validate the slat/double-slot maximum-lift coefficient, landing drag, separated-flow stall or a dynamic flare trajectory.

Table 97. Gear geometry and source-recorded loading states

Parameter	Controlled value
Nose-gear station, xN	FS 0.750 m
Main-gear station, xM	FS 4.290 m
Wheelbase	3.540 m
Main-gear track	3.900 m
CG height used for geometry screen	1.45 m
Forward take-off state	1608.8 kg; FS 3.37374 m
Forward landing state	1519.8 kg; FS 3.35170 m
Actual nominal aft state	1894.8 kg; FS 3.70312 m
U20 10%-SM screening boundary	FS 3.83778 m

9.2 FG8.1 hot-high rotation authority

Table 98. Forward take-off rotation state

Parameter	Value
Mass	1608.8 kg
xCG / moment reference	FS 3.37374 m
Vinf	22.16 m/s
rho	1.079 kg/m ³ for dimensionalization
Mach	0.0634
Re(cbar)	2.41E6
Alpha / Beta	0° / 0°
Fowler	+12°
Aileron / rudder	0° / 0°

Parameter	Value
Propulsion	OFF

$$C_{m,MG} = C_{m,CG} + C_L \frac{x_M - x_{CG}}{\dot{c}} \quad (37)$$

$$M_{aero,MG} = q S \dot{c} C_{m,MG} \quad (38)$$

$$R_N = \frac{W(x_M - x_{CG}) - M_{aero,MG}}{x_M - x_N}, R_N \leq 0 \text{ at unload} \quad (39)$$

Table 99. Executed elevator sweep and sign audit

GUI elevator command	C_L	Cm about CG	Interpretation / use
0°	0.84704	+0.02659	Neutral baseline
+5°	0.91488	-0.18134	Opposite (nose-down moment) direction
+10°	0.98171	-0.38562	Opposite direction
+15°	1.04744	-0.58513	Opposite direction
+20°	1.11211	-0.78034	Opposite direction
-15°	0.64114	+0.66859	Nose-up authority; rotation crossing vicinity
-20°	0.58037	+0.86128	Nose-up authority; converged positive margin

Table 100. Main-gear moment and nose-wheel unload results

Case	Cm,MG	Maero,MG (kN m)	RN (kN)	Convergence disposition
0°	0.4392	6.566	+2.229	Baseline
-15°	0.9809	14.664	-0.059	Authority demonstrated; final-step convergence note
-20°	1.1440	17.103	-0.748	PASS; longitudinal gates satisfied

Table 101. High nose-up elevator convergence audit

FG8.1 convergence item	-15°	-20°	Gate / interpretation
CL 19→20	0.702%	0.199%	<0.5%; -15 NOTE, -20 PASS
CLw 19→20	0.697%	0.064%	<0.5%; -15 NOTE, -20 PASS
CDiw 19→20	0.775%	0.102%	<0.5%; -15 NOTE, -20 PASS
Delta Cm 19→20	0.01432	0.00280	<0.005; -15 NOTE, -20 PASS

Approximately 15° nose-up elevator reaches the rotation crossing, but the -15° GUI case misses the preferred final-step gates. The converged -20° case provides the positive-authority demonstration, with calculated nose reaction -0.748 kN. Negative computed reaction denotes an unload tendency, not a tensile tyre reaction. High-deflection lateral coefficients are not used as lateral-directional validation.

$$|\delta_{e,max,required}) \geq \frac{|\delta_{e,rot})}{0.90} \approx \frac{15^\circ}{0.90} \approx 17^\circ \quad (40)$$

Disposition. The recorded forward take-off case passes the conceptual unpowered rotation screen, with no propwash, powered-lift or thrust-line pitching-moment credit. The approximately 17° value is a design travel requirement with 10% geometric reserve, not a measured hardware stop.

9.3 FG8.2 landing trim and flare authority

Table 102. Landing Fowler-surrogate numerical health

Quantity	Final value	19→20 change	Disposition
C_L	1.59808	0.064%	PASS
$C_{L,w}$	1.63859	0.267%	PASS
$C_{D_i,w}$	0.11955	0.513%	MARGINAL; not used as drag evidence
C_m	+0.06752	0.000264 absolute	PASS
CY / CI / Cn	+0.00317 / +0.00206 / -0.00113	-	Numerical-bias note; not used laterally

Table 103. Forward landing state

Parameter	Value
Mass	1519.8 kg
xCG	FS 3.35170 m
Vinf	21.72 m/s

Parameter	Value
rho	1.079 kg/m ³
Mach	0.0621
Re(cbar)	2.35E6
q	254.51 Pa
Required CL	1.95198
Fowler	+35°
Propulsion	OFF

$$C_{L,req} = \frac{W}{qS} = 1.95198 \quad (41)$$

Table 104. Executed landing trim sweeps

Elevator	Alpha	C_L	C_m
0°	0°	1.59819	+0.06743
0°	4°	1.96032	-0.14137
0°	8°	2.30896	-0.36888
-10°	0°	1.46331	+0.49134
-10°	4°	1.82470	+0.28110
-10°	8°	2.17648	+0.05205

At the required lift, the neutral-elevator sweep gives approximately 3.91° angle of attack and moment -0.13656; the -10° elevator sweep gives 5.45° and +0.19822. The two states bracket zero moment.

$$\delta_{e,trim} \approx -4.08^\circ \text{ (GUI)}, \alpha_{trim} \approx 4.54^\circ \quad (42)$$

Table 105. Interpolated landing trim and demonstrated reserve

FG8.2 result	Value
Required CL	1.95198
Neutral-elevator Cm at required CL	-0.13656
-10° elevator Cm at required CL	+0.19822
Estimated steady trim elevator	~4.08° nose-up
Estimated trim alpha	~4.54°

FG8.2 result	Value
Tested additional elevator beyond trim	~5.92°
Additional nose-up moment at -10°	~2.85 kN m

Disposition. The recorded forward landing state trims at approximately 4.08° nose-up elevator. The tested -10° state provides approximately 5.92° further nose-up movement and about 2.85 kN m additional moment at the required lift. This is a quasi-steady longitudinal authority screen, not a dynamic flare or high-lift validation.

9.4 FG8.3 flap-transition continuity

Table 106. Neutral-elevator flap-transition coefficients — panel A

Alpha	CL @35°	Cm @35°	CL @20°
0°	1.59819	+0.06743	1.11818
4°	1.96032	-0.14137	1.49415
8°	2.30896	-0.36888	1.86278

Table 107. Neutral-elevator flap-transition coefficients — panel B

Alpha	Cm @20°	CL @12°	Cm @12°
0°	+0.03078	0.84699	+0.01670
4°	-0.16386	1.22843	-0.16885
8°	-0.37616	1.60411	-0.37271

Table 108. Flap-transition convergence checks

Configuration	Alpha	CL 19→20	Delta Cm 19→20	CDiW 19→20	Disposition
20°	0°	0.053%	0.00156	0.045%	PASS
20°	4°	0.063%	0.00091	0.635%	Longitudinal PASS; wake-drag note
20°	8°	0.064%	0.00058	0.267%	PASS
12°	0°	0.042%	0.00005	0.029%	PASS
12°	4°	0.039%	0.00018	0.051%	PASS
12°	8°	0.021%	0.00009	0.036%	PASS

At the established landing trim attitude, the constant-required-lift interpolation gives approximately 0.0335 moment coefficient per degree of nose-up elevator. Flap retraction reduces lift monotonically and produces a smooth moment progression with no configuration-induced pitch reversal. The marginal wake-drag point is not performance evidence.

$$\Delta \delta_e \approx - \frac{\Delta C_m}{\partial C_m / \partial \delta_e} \quad (43)$$

Table 109. Approximate flap-transition trim increments

Fowler state	Cm at alpha~4.54°	Delta Cm vs 35°	Approx. additional nose-up elevator	Approx. trim command
35°	-0.17184	0	0	4.08°
20°	-0.19230	-0.02046	~0.61°	~4.69°
12°	-0.19615	-0.02431	~0.73°	~4.81°

Disposition. The 35° → 20° → 12° sequence passes the stated quasi-steady pitch-continuity screen. It does not model propeller transients, flap-rate dynamics, ground effect, pilot response or a powered six-degree-of-freedom go-around. Symmetric power and acceleration are not credited to this result.

9.5 FG8.4 static and landing-impact loads

$$R_N = W \frac{x_M - x_{CG}}{x_M - x_N}, R_{M,total} = W - R_N, R_{M,each} = \frac{R_{M,total}}{2} \quad (44)$$

Table 110. Static gear reactions — panel A

Condition	Mass (kg)	xCG (m)	Nose fraction
Forward take-off	1608.8	3.37374	25.88%
Forward landing	1519.8	3.35170	26.51%

Table 111. Static gear reactions — panel B

Condition	RN (kN)	Main total (kN)	Each main (kN)
Forward take-off	4.084	11.693	5.847
Forward landing	3.950	10.954	5.477

The light-forward cases carry 25.9–26.5% of weight on the nose gear, above the early 10–20% placement guide, but the static reaction is approximately 3.95–4.08 kN. Steering effort, shimmy, tyre loading and braking remain ground-test items; this analytical result does not establish acceptable handling.

$$n_e = 1 + \frac{V_z^2}{2g(\eta_s s_s + \eta_t s_t)} \quad (45)$$

The recorded inputs are vertical speed 5.0 m/s, oleo stroke 0.36 m, tyre deflection 0.13 m, oleo efficiency 0.80 and tyre efficiency 0.58. They give $n_e = 4.508$, reported as 4.51, and ultimate factor $1.5 n_e = 6.761$, reported as 6.76.

$$P_{M,limit} = \frac{n_e W_{LDG}}{2}, P_{M,ultimate} = 1.5 P_{M,limit} \quad (46)$$

Table 112. Landing-gear demand versus existing analytical design basis

Load item	Actual FG8.4 demand	Existing design basis	Assessment
Each main, limit	33.61 kN	44.2 kN	PASS; capacity/demand $\approx$ 1.315
Each main, ultimate	50.41 kN	66.3 kN	PASS; capacity/demand $\approx$ 1.315
Worst forward nose, limit screen	18.42 kN	Supplier rating not frozen	Hardware requirement
Worst forward nose, ultimate screen	27.63 kN	Supplier rating not frozen	Hardware requirement

The main-gear comparison conservatively assigns the entire vertical reaction to the two mains at the recorded 1,519.8 kg landing state. Demand is 33.61 / 50.41 kN per main, below the existing 44.2 / 66.3 kN sizing basis. This is a design-basis comparison, not a supplier-certified capacity. The calculated nose requirement is 18.42 / 27.63 kN; a rounded minimum hardware requirement of 18.5 / 27.7 kN is retained.

9.6 Tip-back and lateral-overturn screen

$$\theta_{TB} = \tan^{-1} \left(\frac{x_M - x_{CG}}{h_{CG}} \right) \quad (47)$$

$$d_{\perp} = \frac{y_M (x_{CG} - x_N)}{\sqrt{(x_M - x_N)^2 + y_M^2}}, \theta_{OT} = \tan^{-1} \left(\frac{d_{\perp}}{h_{CG}} \right) \quad (48)$$

The source geometry screen uses CG height 1.45 m and half-track $y_m = 1.95$ m. The reported overturn angle is defined by the preceding equation; interpretation must use that same geometric convention.

Table 113. Recorded tip-back and overturn geometry

CG state	FS xCG (m)	Tip-back angle	Overturn angle	Disposition
Forward landing	3.35170	32.91°	40.88°	Forward ground-handling reference
Forward take-off	3.37374	32.29°	41.12°	Forward ground-handling reference

CG state	FS xCG (m)	Tip-back angle	Overturn angle	Disposition
Actual nominal aft	3.70312	22.04°	44.50°	PASS; improved vs old aft boundary
U20 aft screening	3.83778	17.32°	45.78°	PASS screening boundary
Obsolete FS 3.900 comparison	3.90000	15.05°	46.35°	Reference only; not operational limit

Disposition. The current gear geometry is retained at conceptual level. Main-gear sizing, forward rotation, landing trim and flap-transition screens close on the recorded cases. Nose-gear hardware, ground handling, high-lift correlation and updated mass-arm confirmation remain qualifications; no late geometry redesign or extra solver run is implied by this document.

10. Validation limits and submission traceability

10.1 Consolidated uncertainty and evidence gaps

Table 114. Open validation register

Domain	Recorded boundary	Evidence required for closure
Aerodynamics	Surface induced drag remains about 5–6% mesh-sensitive; PDT operational/gear drag is model-form dependent	Independent aerodynamic/flight correlation; retain the conservative FDR performance basis
High lift / field performance	FG8 control surrogates do not validate landing maximum lift 3.30 or the narrow adverse landing-distance margin	Configuration-specific high-lift, braking/reverse and field-test correlation
Fuselage / CG	Classical moment correction excludes nacelle/propeller/body-lift terms; all-at-bounds loading fails aft screen	CAD-quality component arms, complete-aircraft validation and explicit uncertainty treatment; incorporate tank increment
OEI control	Physical-stop VMC passes narrowly; 90% simultaneous travel fails; powered-roll allowance is small	Actual control stops; supplier propeller maps; asymmetric powered-lift roll evidence
Dynamic modes	Direct damping signs are favourable; full coupled roots not recorded here	Controlled G1.3 inertia tensor and longitudinal/lateral eigenvalue solutions
Landing gear	Nose-gear supplier rating not frozen; light-forward nose fraction 25.9–26.5%	Hardware substantiation, steering/shimmy/tyre/braking and ground-handling evidence
Mission mass / fuel	FDR mass shortfall 179.9 kg; nominal 190 L volume closes but upper-flow 201.2 L does not	Substantiated mass reduction; installed-engine fuel-flow/fuel-type correlation
Reproducibility	Exact OpenVSP/VSPAERO builds and immutable archive identifiers absent from source	Original environment metadata, run logs, repository commit/tag and ZIP manifest/checksum

10.2 What the evidence does and does not establish

The record establishes a traceable analytical campaign: configuration build-up, convergence studies, repeatability checks, physical-validity rejection decisions, accepted derivative sets and bounded control/gear calculations. It does not establish that the complete aircraft meets every RFP requirement, that supplier hardware is qualified, or that a reviewer can reproduce a run without the referenced native inputs and environment metadata.

The accepted evidence and the open gates must be read together. A closed analysis task means that its method, results and limitation have been recorded; it does not mean the corresponding aircraft-certification question is closed. No AVL, X-Plane, hardware or flight-test result is attributed

to this compendium unless it appears in the source evidence. No X-Plane additional-award claim is made.

10.3 Repository packaging and review route

This compendium is intended to sit beside the analysis ZIP in the GitHub repository. The repository README/release manifest should identify the matching FDR, this compendium, the exact ZIP filename, a release tag and commit identifier, and the ZIP's SHA-256 checksum. The uploaded source log does not supply those identifiers, so this document does not fabricate a link or claim an independently verified archive match.

For each accepted result, the archive should preserve the native geometry/model, run inputs or setup, complete solver history/residuals, raw CSV outputs and the calculation or plotting script/workbook. Diagnostic and rejected runs should remain separately labelled. The test IDs and filenames in Appendix A are the crosswalk into that archive; directory paths and case-sensitive basenames must be checked against the actual ZIP before publication.

Recommended reviewer sequence: read the decision ledger; locate the relevant test ID; inspect its input/reference state and numerical/physical gates; inspect the raw output and post-processing; then compare the reported value and limitation with the FDR. A filename list provides traceability, while the input/output and reproducible calculation provide the evidence.

Appendix A. Evidence-file and folder index

Entries below preserve filenames explicitly recorded in the source. They are references, not a claim that the corresponding files were present in the editorial upload or that their checksums were verified. Ellipses and shorthand remain flagged; a repository manifest must resolve the exact path. No run date or software version is inferred from a filename.

Table 115. Folder-to-evidence crosswalk

Source folder / block	Technical scope	Compendium location
01_CompGeom	Geometry sets, component intersections and exposed area	Section 3.1
02_ParasiteDrag	PDT component build-up and allowances	Sections 3.2–3.3
03_VSPAERO_Convergence	Mesh, wake-node and iteration sensitivity	Section 4.1
04_Clean_Aero	CA-G0/G1/G3 and rejected mixed-body diagnostics	Sections 4.2–4.4
05_Stability	G5 sideslip, rates and control derivatives	Section 5
06_CG_Static_Margin	SM6 reference audit, fuselage correction and CG screen	Section 6
07_VMC_Control_Closure	VMC7.1–3 diagnostics and VMC7.4 equilibrium	Sections 7–8
08 / FG8	Forward control, flap transition and gear closure	Section 9

Table 116. Clean-aero scripts and result files

Group	Files
G1.4 scripts	Tyche_R2T1_G1_4_Analysis.vspscript
G1.5 scripts	Tyche_R2T1_G1_5_Analysis.vspscript
G1.6 scripts	Tyche_R2T1_G1_6_Analysis.vspscript; Tyche_R2T1_G1_6_Analysis_H2.vspscript
G1.4 result CSVs	TRIAL.csv; H1_MODE2_MW_FUSE_FAIRING_A4.csv; H2_Mode_1.csv
G1.5 result CSVs	H2_Mode_1(1).csv; HT_TEST.csv; HT_Test_2.csv; H2_Test_2.csv; H2_NO_VT_TEST.csv
G1.6 result CSVs	HT_TEST_1.csv; HT_TEST_2(1).csv; HT_TEST_3.csv; HT_TEST_4.csv; H2_NO_VT_TEST_1.csv; H2_TEST_1.csv

Group	Files
Clean-aero production evidence	CA_G0_Clean_Lifting.csv; 01_G1_Strut_Sensitivity.csv; CA_G3_Repeatability_A4.csv

Table 117. VMC campaign file audit

Test ID	Controlled source file	Purpose / configuration	Result and use
VMC7.1	VMC7_1_LOWSPD_UNPWR_A0.csv	Alpha 0 degrees, beta 0 degrees, unpowered, low-speed baseline, 20 iterations	PASS; reference baseline
VMC7.1 duplicate	VMC7_1_LOWSPD_UNPWR_A0(1).csv	Same timestamp and byte content as VMC7.1	Archive duplicate; no new test
VMC7.2 initial	VMC7_2_TOFLAP12_ALPHA_0_4_8.csv	Unpowered alpha 0/4/8 degrees; intended Fowler +12 degrees	REJECTED; asymmetric flap grouping produced spurious roll
VMC7.2 corrected	VMC7_2_TOFLAP12_ALPHA_0_4_8(1).csv	Corrected symmetric Fowler +12 degrees; alpha 0/4/8 degrees	PASS for flap implementation and low-speed trend
VMC7.3A initial	VMC7_3_SYM_TOFLAP12_ALPHA_0_4_8.csv	First powered symmetric alpha sweep	REJECTED; flap-grouping contamination and non-physical disk input basis
VMC7.3B corrected grouping	VMC7_3_SYM_TOFLAP12_ALPHA_0_4_8(1).csv	Corrected Fowler grouping, powered alpha 0/4/8 degrees, 20 iterations	Development evidence only; powered oscillation and old $C_T = 0.40$ basis
VMC7.3C	VMC7_3C_SYM_A0_40IT.csv	Corrected C_T / C_p , co-rotating, relaxation 1.0, 40 iterations	FAIL production convergence
VMC7.3D	VMC7_3D_SYM_A0_RLX05.csv	Same as VMC7.3C, relaxation 0.5	Improved some spans; still FAIL
VMC7.3E	VMC7_3E_CR_SYM_A0_RLX05.csv	Counter-rotating, relaxation 0.5	Physical symmetry improved; numerical FAIL
VMC7.3F	VMC7_3F_CR_SYM_A0_RLX025.csv	Counter-rotating, relaxation 0.25, 40 iterations	Final numerical test; FAIL and close powered branch

Test ID	Controlled source file	Purpose / configuration	Result and use
VMC7.4	Deterministic hybrid calculation in this record	G5 derivatives + explicit OEI force/moment model + finite controls	CONDITIONAL ANALYTICAL PASS / RELEASE HOLD

Table 118. FG8 executed CSV register

Test	File(s)	Use in this log
FG8.1 baseline / positive commands	FG8_1_HH_ROT_DE0.csv; FG8_1_HH_ROT_DE_M5.csv; ..._M10.csv; ..._M15.csv; ..._M20.csv	Elevator sign audit and opposite-direction response
FG8.1 nose-up authority	FG8_1_HH_ROT_DE_M15(1).csv; FG8_1_HH_ROT_DE_M20(1).csv	Main-wheel rotation closure
FG8.2 sanity	FG8_2_LDG35_SANITY_A0_DE0.csv	+35° Fowler numerical health
FG8.2 approach trim	FG8_2_LDG35_DE0_A0_4_8.csv	Neutral-elevator lift/pitch surface
FG8.2 flare authority	FG8_2_LDG35_DE_NEG10_A0_4_8.csv	Nose-up elevator trim bracket and reserve
FG8.3 transition 20°	FG8_3_GA20_DE0_A0_4_8.csv	Intermediate flap-transition state
FG8.3 transition 12°	FG8_3_GA12_DE0_A0_4_8.csv	Released take-off/go-around flap state

Table 119. Analytical evidence and provenance register

Evidence item	Source / calculation basis	Use
FG8.4 gear closure	SM6.4 forward/aft CG states; xN=0.750 m; xM=4.290 m; 4.51 g / 6.76 g released gear basis	Static reactions, main/nose design loads, tip-back and overturn
VMC atmosphere repair	Converged G5 derivatives; VMC-A/B/C; P=160.13 kW; y=1.60 m; CD,f bounds; finite controls; explicit p=1.225 and 1.079 cases	Supersedes ambiguous p=1.168 VMC headline values
Parasite-drag uncertainty	G2_COMPLETE_AIRFRAME_ParasiteDrag.csv; G2_C1_Operational_ParasiteDrag.csv; G2_C2_UpperBound_ParasiteDrag.csv	Geometry/operational/upper PDT band and conservative report treatment

Evidence item	Source / calculation basis	Use
CG / neutral-point uncertainty	Tyche_R2T1_SM6_3B_SM6_4_Integrated_CG_Closure_Pre_VMC record	U11/U14/S14/U20 NP sensitivity; loading-cloud and arm uncertainty
Powered-flow model boundary	VMC7_3C/D/E/F CSVs and 07_VMC_Control_Closure analysis log	Documents non-convergent powered free-wake branch and counter-rotation sensitivity
Folder 08 VSPAERO evidence	FG8_1, FG8_2 and FG8_3 CSV files listed in Appendix A	Forward rotation, landing trim/flare and go-around transition

Table 120. Additional explicitly recorded G5 files

Test	Filename / source identifier
B0.1	G5_B0_REPEAT_A4.csv
B0.2	G5_B0_A0_B0.csv
SM6 integrated closure	Tyche_R2T1_SM6_3B_SM6_4_Integrated_CG_Closure_Pre_VMC record; exact file extension/path not specified in source

Filename qualification. The FG8 positive-command “M5 / M10 / M15 / M20” naming does not establish physical nose-up direction. Use the recorded control-group signs and executed GUI commands. The source abbreviates some FG8 filenames with ellipses and some powered filenames with C/D/E/F shorthand; those entries must not be mistaken for literal complete archive paths.

Appendix B. Superseded results retained for audit

This appendix preserves the main superseded numerical branches as historical evidence. They are not current released values. Their replacement is a documented engineering decision, not deletion of an unfavourable result.

Table 121. Historical result and governing replacement

Historical branch	Retained historical result	Why superseded / governing replacement
G2 strut-gap geometry	141.533 m ² exposed area; 13,376 triangles	Corrected attachments: 141.591 m ² ; 13,888 triangles
Uncorrected lifting-system stability	FS 4.14286 m NP; 12.91% margin at FS 3.900 m	Fuselage correction gives central NP 4.02894 m and only 6.86% at FS 3.900; U20 aft screen 3.83778 m
SM6.2 planning note	Reference-shift audit initially described as unnecessary	Audit subsequently executed; matched NP difference 0.47 mm
Early SM6.4 evidence gap	Component mass arms unavailable; loading-cloud closure HOLD in that issue	Later integrated closure and case summaries used by VMC/FG8; underlying ledger and updated tank arm remain required
VMC mixed-density full travel	VMC-A 38.73 kn windmill / 39.09 kn severe	Explicit sea-level atmosphere gives 38.97 / 39.37 kn EAS
VMC mixed-density 90% travel	VMC-A approximately 40.73–40.74 kn windmill in source rounding	Corrected sea-level result 40.97 kn EAS; robustness remains FAIL
VMC extra powered roll	Approximately 0.00250 allowable adverse coefficient at 39.96 kn	Corrected windmill / severe allowances approximately 0.00206 / 0.00122
Early mass/fuel closure	OEW 1,327.0 kg; fuel 109.8 kg; full mission 2,152.8 kg	FDR rebase: OEW 1,329.0 kg; nominal fuel 134.9 kg; full mission 2,179.9 kg

Table 122. Historical mixed-density VMC full-travel results — superseded

CG case	$C_{D,f}=0.08$	0.25	0.45 windmill	0.60 severe
VMC-A	37.83 kn	38.22 kn	38.73 kn	39.09 kn
VMC-B	37.69 kn	38.08 kn	38.56 kn	38.96 kn
VMC-C	37.67 kn	38.03 kn	38.54 kn	38.95 kn

Table 123. Historical mixed-density VMC 90%-travel results — superseded

CG case	$C_{D,f}=0.08$	0.25	0.45 windmill	0.60 severe
VMC-A	39.62 kn	40.14 kn	40.73 kn	41.15 kn

Table 124. Historical powered-roll sensitivity — superseded atmosphere

Added adverse $ C_{l,pw} $	Calculated VMC-A, $C_{D,f}=0.45$
0	38.73 kn
0.001	39.20 kn
0.002	39.70 kn
0.0025	approximately 39.96 kn
0.003	40.21 kn
0.005	41.29 kn
0.010	44.49 kn

Table 125. Early mass cases — superseded fuel and mass basis

Case	OEW kg	Payload kg	Fuel kg	Total kg	CG evidence
Operating empty	1327.0	0	0	1327.0	OEW arm absent; CG unknown
505 km MTOM case	1327.0	563.2	109.8	2000.0	Payload/fuel arms absent
Full RFP arithmetic	1327.0	716.0	109.8	2152.8	Mass FAIL; not flyable
Full payload, zero fuel	1327.0	716.0	0	2043.0	Mass FAIL; not flyable
Hot-high landing limit	-	-	-	1900.0	Loading distribution absent

The earlier atmosphere used an incompletely traced 1.168 kg/m³ thrust density with EAS-referenced airframe loads. Its equilibrium method was retained, but the bookkeeping was corrected. None of the historical results above is used as the compendium's governing VMC, mass or operational CG conclusion.

B.1 Initial geometry-set and mode definitions

The original log also contains an embedded image of the initial set/mode register. Its content is preserved below as editable tables. These are source configuration definitions, not evidence that every intended mode produced an accepted solution. The later thin-model selection, powered-branch rejection and FG8 control-surrogate limitations govern actual use.

Table 126. Initial geometry-set organisation — historical source register

Set	Included	Excluded / purpose
G0_BARE_AIRFRAME	Fuselage, wing, HT, VT	Nacelles, propellers, struts and gear excluded; bare-airframe aerodynamics
G1_NACELLE_AIRFRAME	G0 + both nacelles / spinners	Propellers, struts and gear excluded; nacelle drag penalty
G2_COMPLETE_AIRFRAME	G1 + struts + fixed landing gear	Propellers excluded; complete unpowered aircraft
G3_POWERED_AIRCRAFT	G2 + both propellers / disks	Intended powered-wing and asymmetric-propulsion analysis; later powered coefficients rejected

Table 127. Initial mode definitions — not a released aerodynamic database

Mode / set	Slat / Fowler	Propeller state	Original purpose
M_CRZ / G2	0° / 0°	Excluded	Clean cruise / stability
M_TOU / G2	15° / 12°	Excluded / OFF	Unpowered take-off high lift
M_TOP / G3	15° / 12°	Symmetric powered disks	Powered take-off
M_LDG / G2	25° / 35°	No lift credit	Landing
M_VMC / G3	15° / 12°	One live; one failed / feathered	Critical OEI control

Appendix C. Source notes and editorial control**Table 128. Source and attribution register**

Source	Role
Tyche R2-T1 Analysis and Validation Evidence Log.odt	Primary source for recorded geometry, numerical runs, coefficients, analytical derivations, filenames and dispositions
Tyche_R2-T1_Final_Design_Report_G1_3_FINAL_Revised(3).pdf	Controlling report context for the R2-T1 / G1.3 / S1.0 / CVR-1.0 state; not edited by this compendium
Gudmundsson, General Aviation Aircraft Design, Appendix C1	Munk–Multhopp source implementation attributed in the original log; degree/radian conversion and forward/aft branch equations retained
RAeS Light Aircraft Design Competition 2025–26 rules	Competition context; this supporting record is not a substitute for the report's stated requirements and evidence

Editorial changes are limited to submission packaging and evidence clarity: new title/front matter, tool register, contents and pagination; formal topic-based organisation; consolidation of duplicate material; decision-led rejection records; restoration of equation typography; clear supersession and model-use boundaries; and an archive crosswalk. Numerical results remain attributed to their recorded run or calculation basis. No missing version, solver run, source file, hardware result or repository identity has been invented.